When Death Was Postponed: The Effect of HIV Medication on Work, Savings, and Marriage *

Mette Ejrnæs† Esteban García-Miralles‡
Mette Gørtz§ Petter Lundborg¶

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Abstract

Increased life expectancy can affect individuals' incentives to work, save, and marry, independent of changes in their underlying health. To test this hypothesis, we leverage the sudden introduction of a groundbreaking treatment in 1995, which significantly extended the life expectancy of HIV-infected individuals. Our analysis compares the behavioral responses of HIV-infected individuals who were still in good health but who differed in their access to the new treatment. Those with access to treatment work substantially more, marry later, but do not save more. These findings underscore the importance of accounting for incentive effects when assessing the value of improvements in life expectancy.

Keywords: Life Expectancy, Labor Supply, Marriage, HIV
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†University of Copenhagen and CEBI. (email: mette.ejrnes@econ.ku.dk)
‡Banco de España. (email: esteban.garcia.miralles@bde.es)
§University of Copenhagen, CEBI, and IZA. (email: mette.gortz@econ.ku.dk)
¶IZA and Department of Economics, Lund University. (email: petter.lundborg@nek.lu.se)

People who had been planning to die sooner rather than later – quitting their jobs, cashing in their insurance policies, running their credit cards to the limit, avoiding fresh romances or clinging to old relationships – began finding themselves back in the business of living, with all its complications.

"*From the AIDS Conference, Talk of Life, not Death*" by David W. Dunlap. Published in New York Times, July 15, 1996.

1 Introduction

In the last century, global life expectancy has shown a remarkable increase, owing to factors such as rising incomes, widespread health insurance, and advancements in medical treatments. Human capital theory posits that a longer life expectancy strengthens the incentives to invest in skill acquisition (Becker, 1964; Ben-Porath, 1967), forming the foundation for theories that link life expectancy to economic growth through a human capital channel (Kalemli-Ozcan, 2002; Soares, 2005; Murphy and Topel, 2006; Weil, 2007). A longer life expectancy has implications that extend beyond the realm of human capital theory, however, since major economic decisions about labor supply, savings, and marriage, are also taken under uncertainties about one's (remaining) life-span.[1] With a longer planning horizon, individuals' incentives regarding work-leisure trade-offs, retirement savings, and marriage market behavior are affected, with significant implications for both welfare and economic growth.

As longer lifespans are commonly linked to better health, often resulting from medical innovations, it has proven challenging to empirically disentangle the pure incentive effect of a longer life expectancy from the effects of improved health. Healthier workers are more productive and improvements in health can lead to better employment opportunities, a prolonged working life (Dobkin et al., 2018; Fadlon and Nielsen, 2021; Stephens and Toohey, 2022), and improved prospects in the marriage

[1] See discussions in Blundell and MaCurdy (1999); Browning and Crossley (2001); Dynan et al. (2002, 2004); Cocco and Gomes (2012); Haan and Prowse (2014); Blundell and Macurdy (2017); Low et al. (2018).

market (Chiappori, 2020).[2] Yet, understanding the role of the incentive effect is crucial for valuing increases in life expectancy in itself and for valuing the benefits of new medical technologies, as cost-benefit assessments are likely to underestimate these benefits if the pure incentive effects of a longer life expectancy are not properly accounted for. More generally, such incentive effects may arise whenever individuals form expectations about future health improvements under uncertain access to treatment.

In this paper, we estimate the incentive effect of greater life expectancy on labor market outcomes, financial outcomes, and marriage market outcomes. We do so by focusing on a medical breakthrough that led to a sudden and dramatic increase in life expectancy for HIV-infected individuals. The treatment, known as Highly Active Antiretroviral Therapy (HAART), was gradually introduced through 1995 and formally approved in 1996, and it significantly improved survival probabilities among HIV-infected individuals (Legarth et al., 2014; Mocroft and Lundgren, 2004). Before the introduction of HAART, an HIV diagnosis was associated with a substantially shortened life expectancy. However, with the advent of HAART, HIV was transformed into a manageable chronic illness.[3]

Specifically, we define our treatment group as individuals diagnosed with HIV after the introduction of HAART, who therefore knew about the new medication when they were diagnosed. Our control group consists of individuals diagnosed before HAART medication was discovered, and are kept in our analysis only up to 1994, before they learn about the new medication. Therefore, although both groups experience the shock of being diagnosed with HIV, the treatment group can anticipate a significantly enhanced life expectancy at the time of diagnosis compared to the control group due to the availability of HAART.[4,]

[2]In addition, the marginal utility of consumption may increase with better health since many consumption goods, such as travel, are complements to good health (Finkelstein et al., 2013).

[3]Prior to 1995, numerous clinical trials were conducted on anti-retroviral drugs, primarily involving HIV patients who were already in a relatively advanced stage of the disease. However, the outcomes of these trials had not yet demonstrated significant success (Hamilton et al., 2021). Papageorge (2016) shows a significant increase in hopefulness about the future among HIV patients since 1995 due to the introduction of HAART.

[4]As we show in Section 2.3, before the introduction of HAART, 80% of all individuals diagnosed

In our analyses, we utilize unique and high-quality longitudinal register data from Denmark on HIV-infected individuals, observed before and after they receive their HIV diagnosis. These data help us to address two main empirical challenges. First, they help in isolating the incentive effect of greater life expectancy from the effect on contemporaneous health. As the data include clinical information on the disease stage, measured through blood samples and CD4 cell counts, we are able to select and study individuals who, at the time of the HIV diagnosis, were still in good health and would not experience any health symptoms in the following four to five years.[5] By comparing changes in outcomes before and after the HIV diagnosis across the two groups –those diagnosed before and those diagnosed after the introduction of HAART–, we can estimate the incentive effect of longer life expectancy, net of any changes in underlying physical health. While the introduction of HAART might have changed incentives to get tested for HIV, we show that the number of diagnoses for the two groups was very similar and stable over time, and that their average CD4 count level (and its distribution) at the time of diagnosis was remarkably similar.

Second, as we are comparing changes in the outcomes of HIV-infected individuals over different periods of time, it is crucial to address potential confounding factors related to calendar time, such as business cycle, structural changes, or reforms. To account for such changes, we use data on the full population of Denmark and separately match a control group of non-infected (HIV–) individuals to the HIV-infected (HIV+) individuals based on their cohort, age, and gender. Our empirical analyses thus effectively correspond to a triple-difference design.

Our first set of results shows that the introduction of HAART and the subsequent increase in life expectancy reduced the negative effect of receiving an HIV diagnosis on labor supply and earnings. In the four years after diagnosis, when health had not yet deteriorated, access to HAART raised employment by 8 percent and earn-

with HIV died within 4 to 5 years after diagnosis. In our analyses, however, we focus on a sample of individuals who were diagnosed before the disease has progressed much. For them, life expectancy at the time of diagnosis is estimated at around 16 years, as further described in Section 3.7.

[5]We show that individuals in this sample, regardless of their access to HAART, are equally likely to remain above healthy levels of CD4 cell counts, exhibit equally low rates of infections and mental health diagnoses, and experience similar low mortality rates during the 4-5 year period following their HIV diagnosis. It should also be noted that we exclude drug addicts from our analyses.

ings by 9 percent among those who remained employed. These effects are largely driven by sharp declines in employment and earnings among those diagnosed before 1995, who experienced a significant drop in life expectancy, suggesting a substitution toward leisure as life expectancy declined. Importantly, the observed effects reflect adjustments at both the extensive and intensive margins. The higher employment rate indicates a greater likelihood of remaining in the labor force (extensive margin), while the earnings increase suggests that those who remained employed also worked more hours or transitioned to higher-paying jobs (intensive margin).

Our second set of results reveals that the increase in life expectancy resulting from HAART had limited effects on financial decisions, including bank account savings, stock market participation, and home ownership. We interpret these results in the light of standard life-cycle theory; on the one hand, theory predicts that wealth is depleted as life expectancy is reduced following a health shock, but, on the other hand, a sharp reduction in life expectancy in the context of an HIV diagnosis could lead to increased precautionary savings in response to increased uncertainty (Davies, 1981; Kotlikoff, 1989). Furthermore, the impact of increased life expectancy on saving towards retirement may be limited in countries with generous pension systems and widespread passive savings behavior (Chetty et al., 2014; García-Miralles and Leganza, 2024b) and in the presence of strong bequest motives (Hurd, 1987; Dynan et al., 2002; De Nardi et al., 2009).

Our third set of results suggests that HAART had large effects on marriage and cohabitation rates. In the group of HIV+ individuals diagnosed before 1995, who faced a much-reduced life expectancy, marriage rates went up after being diagnosed. This finding can be interpreted in a family economics framework, where cohabitation and marriage provide important sources of private insurance against health shocks (Anderberg, 2007; Persson, 2020; Potoms and Rosenberg, 2021). In the absence of HAART, the insurance value of having a partner who can provide financial and practical caregiving support significantly increased in light of the large negative shock to life expectancy that followed an HIV diagnosis. Moreover, if relative preferences for consumption and leisure change such that more weight is put on leisure relative to consumption, as the results above suggest, the utility of being in a couple could also increase if leisure complementarities are positive (Lalive and Parrotta, 2017; Browning et al., 2020; Johnsen et al., 2022; García-Miralles and Leganza, 2024a; Georges-Kot et al., 2024).[6]

While our findings highlight significant changes in economic behavior driven by the incentive effect, the improvement in life expectancy due to HAART may also affect mental health, potentially influencing economic behaviors. However, our analysis reveals little evidence of substantial mental health effects of increased life expectancy, aligning with the results obtained by Oster et al. (2013b). Since our mental health measures are limited to visits to psychiatrists and psychologists, however, we are unable to capture all dimensions of mental health, and it therefore remains a possibility that our results reflect both an incentive effect and some impact of improved mental health.

Our results for earnings and labor supply illustrate the importance of accounting for the incentive effect of greater life expectancy when valuing the gains from new medical technologies. While traditional cost-effectiveness studies typically account for only the direct health and socioeconomic effects following new medical treatments (e.g., Davies et al., 1999), our findings pinpoint that the pure anticipation of access to a new type of health technology in the future may have additional positive effects on economic outcomes, even before health starts to deteriorate. Such gains are of particular relevance for diseases with long time lags between receiving information about the disease and the outbreak of health symptoms. For such diseases, the incentive effect may constitute a non-negligible fraction of the total value of life-extending medical technology. We illustrate this phenomenon with a back-of-the-envelope exercise where we show that the incentive effect of HAART on employment may constitute as much as 21 percent of the total effect on employment during the first 15 years after an HIV diagnosis.

Complementing our average results, heterogeneity analyses by age, gender, and education illuminate underlying mechanisms. Responses are similar across age groups despite life-cycle theory predicting stronger effects for the young, suggesting that

[6] Also, experiencing a shorter time period to find a partner and enjoying the benefits of partnership implies that the "option value of waiting" for a high-quality partner declines (Strobel, 2003).

short-run insurance needs, survival uncertainty, and generous public support dampen age gradients. Women show larger (though not statistically significant) effects on savings, assets, and partnership—consistent with stronger insurance motives—but smaller effects on earnings, indicating more limited scope to adjust labor supply. Lower-educated individuals display significantly stronger partnership effects, reflecting greater reliance on a partner for economic and care support. These patterns point to insurance and care considerations as central drivers of behavioral adjustments to longevity gains.

More generally, our findings suggest that behavioral responses to changes in life expectancy occur well before any major health deterioration, a pattern that may extend beyond HIV to other chronic diseases where treatment access is uncertain. In many medical contexts, individuals make forward-looking economic and life decisions in anticipation of potential improvements in health, rather than solely in response to current health status. This insight is particularly relevant in cases where access to new treatments depends on factors such as cost, eligibility criteria, or insurance coverage, which could shape labor supply, savings behavior, and family formation in ways similar to what we document in this paper. While our study focuses on younger individuals, similar incentive effects may arise among older individuals adjusting retirement timing, wealth accumulation, and long-term care planning in response to longevity gains. This highlights the need to account for both incentive effects and direct health effects when evaluating the broader economic impact of medical innovations.

Our paper contributes to several strands of literature. It is most closely related to the small literature that studies how increases in life expectancy—holding health constant—causally affect incentives to save and work (Baranov et al., 2015; Baranov and Kohler, 2018; Papageorge et al., 2021). We provide new evidence from a Western, high-income setting with universal health care, where access to life-extending treatment is not shaped by insurance coverage or out-of-pocket costs, and where administrative register data allow us to study a comprehensive set of economic and family outcomes. This institutional and empirical context differs markedly from prior studies and allows us to analyze the incentive effects of longer life expectancy in a universal-coverage environment, leveraging population-scale administrative data to trace impacts across labor supply (extensive and intensive margins), financial behavior, and partnership decisions.

We complement Baranov et al. (2015) and Baranov and Kohler (2018), who study Malawi and show that increased HAART availability raises savings and farm work among HIV-negative individuals. Our focus on HIV-positive individuals allows us to examine responses to a much larger and more certain increase in life expectancy, while our developed-country setting —with its public pension system— provides a natural contrast to Malawi in understanding why savings responses may differ.[7] We also complement Papageorge et al. (2021), who analyze domestic violence, drug use, and employment among low-income U.S. women with high rates of substance use. In contrast, we study a broader, more representative HIV-positive population of both men and women, exclude drug addicts, and cover a wider range of outcomes—including both extensive and intensive labor supply margins, bank and stock holdings, housing, and partnership formation and quality.[8,9] Methodologically, we go beyond the descriptive analysis of Legarth et al. (2014) by combining synthetic controls with difference-in-differences estimation to credibly estimate causal effects. To our knowledge, we are also the first to study how longer life expectancy affects the incentives for marriage and cohabitation.

Our paper further contributes to the literature on the incentive effects of longer life expectancy on human capital investment (Fortson, 2011; Jayachandran and Lleras-

[7]While Baranov et al. (2015) and Baranov and Kohler (2018) found that HIV-*negative* individuals reacted to the introduction of HAART, such a reaction is much less likely in the Danish context, where the risk of infection was considerably lower. If anything, such a reaction would lead to a somewhat smaller incentive effect, as the difference in life expectancy between HIV-positive and HIV-negative individuals would be attenuated.

[8]In a contemporaneous working paper, Karparti (2022) studies the effect of life expectancy on wealth accumulation, using genetic testing data from a group of individuals at high risk of developing hereditary cancer syndrome.

[9]Our paper also relates to the literature that examines the direct effect of HAART treatment on health, labor market outcomes, well-being, and economic growth (Papageorge, 2016; Thirumurthy et al., 2008; Habyarimana et al., 2010; Thirumurthy and Zivin, 2012; Tompsett, 2020; Da Costa, 2023; Tompsett, 2020). Although these studies typically find large effects of HAART treatment, they do not aim to disentangle the health effects of the treatment from the incentive effects of longer life expectancy.

Muney, 2009; Oster et al., 2013b).[10] While this work has highlighted human capital accumulation, particularly education, as a key response to longer life expectancy among younger cohorts, our study focuses on somewhat older individuals. This allows us to show that longevity gains can alter behavior even when education is no longer a relevant margin, with effects materializing instead through labor supply, financial, and partnership decisions. Our results, together with those of Baranov et al. (2015), Baranov and Kohler (2018), and Papageorge et al. (2021), suggest that focusing solely on human capital accumulation underestimates the broader incentive effects of longer life expectancy. By exploiting a sharp and well-defined medical breakthrough such as HAART, we provide evidence that speaks directly to settings where life expectancy changes due to medical innovations or health policies that alter individuals' perceptions of their longevity.

The paper unfolds as follows. In Section 2, we provide a brief overview of the institutional context of HIV and HIV treatment in Denmark, and we discuss how HAART dramatically changed the situation for HIV-positive individuals. In Section 3, we describe the data sources used in our analyses. Section 4 outlines our empirical approach. We present our main results in Section 5, and Section 5.5 provides a set of robustness analyses. Finally, Section 6 concludes.

2 Background and institutional context

2.1 HIV and AIDS: Medical facts

HIV (Human Immunodeficiency Virus) is a chronic virus that impairs the immune system's ability to defend against ordinary infections, resulting in immunodeficiency. HIV is primarily transmitted through sexual contact and exposure to infected blood.

[10]Fortson (2011) found that regions in sub-Saharan Africa with higher HIV prevalence experienced relatively larger declines in schooling. Jayachandran and Lleras-Muney (2009) found that a sudden drop in maternal mortality in Sri Lanka in the 1950s, which sharply increased the life expectancy of girls, led to an increase in girls' education relative to that of boys. Oster et al. (2013a,b) estimate the impact of life expectancy on human capital investment using data on individuals at risk for Huntington's disease.

The HIV virus attacks the immune system and especially the CD4 cells (T-cells). In the absence of treatment, HIV-infected individuals experience a progression of immunodeficiency, leading to the development of opportunistic infections and, ultimately, AIDS-related mortality.

To monitor the progress of the HIV disease, CD4 counts, defined as the number of white blood cells per mm^3 of blood, are measured by a blood test conducted by a health professional. Without HIV, a healthy immune system has a CD4 count between 500 and 1,600 cells per cubic millimeter of blood (cells/mm3). When the CD4 count is below 200 cells/mm3, a person will receive a diagnosis of AIDS. With a cell count above 350, an HIV-positive individual has yet to experience any physical symptoms.

As discussed in the introduction, focusing on this particular subset of individuals whose CD4 counts are high constitutes an integral part of our empirical strategy, since physical symptoms cannot explain any differences in the behavioral responses to new HIV treatments within this group. We provide descriptive results based on our data that support these medical facts, including evidence on health symptoms picking up only for low CD4 counts (Section 3.3), employment falling more severely at low CD4 counts (Section 5.7) as well as a wide set of tests specific to our sample of analysis throughout the paper.

2.2 HIV in society

The first documented scientific report of HIV dates back to 1981, after which the number of cases increased dramatically throughout the 1980s worldwide. In many countries, the HIV/AIDS epidemic represented a significant demographic and economic shock (Karlsson and Pichler, 2015). To date, the disease has claimed the lives of over 32 million individuals across the globe. Prevalence is higher among homosexual men, people with hemophilia, drug addicts, and individuals from sub-Saharan Africa.

In Denmark, the first reports about what is now recognized as HIV/AIDS appeared in Danish newspapers in late 1981. At that time, the disease was believed to be a type of cancer that primarily affected homosexual men. HIV testing became available in 1985, with individuals being able to obtain a blood test free of charge from their general practitioner (GP), at a hospital, or at a clinic for sexually transmitted diseases. Some hospitals and clinics also offered anonymous testing options. The high mortality rate associated with AIDS quickly became apparent in Denmark. From the 1980s until the mid-1990s, AIDS claimed the lives of 175-240 individuals annually in Denmark, corresponding to a mortality rate of 3 out of 100,000 individuals.

2.3 A medical breakthrough: HAART medication

The discovery of Highly Active Antiretroviral Therapy (HAART) medication in 1995 and its introduction to Danish patients in 1996 yielded rapid and substantial reductions in the number of individuals suffering from HIV/AIDS, with mortality rates plummeting to approximately one-third of their previous level. The breakthrough resulted from the combination of three anti-retroviral drugs, including a novel type of medication known as protease inhibitors. In June 1995, the Food and Drug Administration (FDA) approved the first protease inhibitors for the treatment of HIV patients, while the combination therapy was granted approval in December of that same year.[11] About six months later, in July 1996, the promising results of the new combined treatment were confirmed at the XI International AIDS Conference.

The media quickly disseminated information about these medical innovations to the general public. Anecdotal evidence and news articles from Denmark suggest that a sense of optimism began to emerge already in September 1995, with growing awareness that this marked the beginning of a new era with an effective treatment for HIV available (see Appendix B). Papageorge (2016) provides evidence of improved optimism in HIV patients following the introduction of HAART.

Anti-retroviral treatment works by inhibiting some of HIV's enzymes, reducing HIV in the body, and increasing CD4 counts. While the treatment does not cure HIV entirely, it will halt its progression, leading to a significantly reduced risk of

[11] See https://hivinfo.nih.gov/understanding-hiv/fact-sheets/fda-approved-hiv-medicines.

developing and dying from AIDS. In addition, by suppressing viral load, HAART greatly reduces the risk of HIV transmission. Most treatment guidelines, including those in Denmark, recommend initiating HAART treatment when CD4 counts fall below 350.

HAART has transformed HIV into a chronic infection in the Western world, with survival rates similar to those of the general population. Figure 1 shows the survival rates by year following HIV diagnosis in Denmark. Patients diagnosed between 1990 and 1993 had low survival rates, with only 20 percent surviving five years after diagnosis. For patients diagnosed in 1994 or 1995, however, survival curves are much less steep, and at least 50 percent survived five years after the diagnosis. For patients diagnosed after 1996, 80 percent were still alive 5 years after diagnosis.[12]

A study of the living conditions of those infected with HIV in Denmark points to remarkable changes in expectations and hopes for the future for patients diagnosed before and after the arrival of the HAART medication (Carstensen and Dahl, 2007). Patients diagnosed in the early period were more likely to discontinue their education or to report HIV infection as a reason for retirement than patients diagnosed in more recent years.

3 Data and analysis sample

Our study draws upon a novel data source that integrates longitudinal register data for the entire population from Statistics Denmark with a medical database containing clinical health information on individuals diagnosed with HIV. This section outlines the key variables and provides a detailed description of the data.

[12] Note that Figure 1 shows survival curves for all individuals diagnosed with HIV, irrespective of their CD4 count at the time of HIV diagnosis. In our analyses, we instead focus on individuals whose CD4 count levels are equal to or above 400, who face less steep survival curves. In Section 3.7, we discuss the perceived life expectancy at the time of diagnosis for these individuals, which we estimate to be around 16 years.

3.1 Danish Registers

The data from Statistics Denmark allow us to follow the population of HIV-positive individuals from the 1980s until 2000. The register data provides background information as well as socioeconomic outcomes and information on health care use. The socioeconomic outcomes studied in this paper include employment, income, savings, financial market participation, and housing. We deflate all monetary values to 1997 levels. The exchange rate in 1997 was approximately 10 Danish kroner (DKK) to 1.52 U.S. dollars (USD). In addition, the register data provides detailed information on cohabitation and marriage patterns. Statistics Denmark defines cohabitation as two adults who are not family-related, living at the same address, of opposite sexes, and with less than 15 years of age difference.[13] Finally, we utilize Statistics Denmark's healthcare registers to assess the general health status of the individuals before and after being diagnosed with HIV.

3.2 HIV Medical Database

To supplement our register data on socioeconomic characteristics and healthcare use, we obtained access to DANHIV, a comprehensive clinical database compiled by all public and private hospitals in Denmark since the 1990s.[14] The data in DANHIV covers all patients diagnosed with HIV (ICD-10 codes B20-24) who were alive in 1995 when the database was initiated. It is important to note, however, that the database also contains retrospective information on the date of HIV diagnosis for those diagnosed before 1995 and the source of infection. Furthermore, each individual in DANHIV has been linked to Statistics Denmark's anonymized register data. Four variables from DANHIV are of particular importance for our analysis:

[13] Unlike cohabitation among people of opposite sexes, there is no information in Statistics Denmark's registers on cohabitation between people of the same sex. Since a significant portion of our sample comprises homosexual men, we are also interested in studying partnering among men. To this end, we use information on registered partnerships, which were introduced in Denmark in 1989 as a legal equivalent to marriage for same-sex couples.

[14] DANHIV is described in Obel et al. (2008). More information on DANHIV (in Danish): https://www.rkkp.dk/kvalitetsdatabaser/databaser/Dansk-HIV-database-/.

Time of diagnosis of HIV and HAART treatment. The DANHIV database provides information on both the date of HIV diagnosis and the date when HAART treatment commenced.

Source of Infection. The medical database contains information on the source of infection, which is self-reported by the patient. The source of infection can be classified as sexual transmission, including heterosexual transmission or transmission by men having sex with men (MSM), or non-sexual transmission such as drug abuse, transmission from mother to child during pregnancy, blood transfusions, or unknown. This information is crucial for defining our sample as we focus on individuals who were infected through the two most prevalent sources of infection: heterosexual contact and men having sex with men. We also use this information to exclude individuals who were infected through drug use from our sample and to examine whether the effects of our interventions vary by sexual orientation.

CD4 Counts. This variable is the leading indicator of immune system health and provides information about the progression of the HIV disease. It is a key variable for defining our sample, which we restrict to healthy HIV-positive individuals whose immune system is not deteriorated at the time of diagnosis nor during the years to follow.

On average, individuals have their CD4 counts measured 2.5 times per year in our sample. To ensure compatibility with our annual outcome measures, we construct an annual CD4 count measure for each individual by taking the mean of all their CD4 count measurements in a given year. It should be noted that HIV patients are regularly informed of their CD4 counts during their visits to the doctor, as this is a critical indicator of their current health status.

The HIV medical database only includes CD4 counts from 1995 onwards. For individuals diagnosed before 1995, we therefore impute their pre-1995 CD4 counts based on the counts observed since 1995. It is important to note that these imputed values are not included as a variable in any of the main regression analyses, but are used solely to identify individuals with sufficient health status at the time of diagnosis (those diagnosed when their CD4 count is above a certain threshold).

Our imputation method estimates a quadratic model with individual fixed effects, with CD4 counts as a function of time from diagnosis. We estimate this model using observations between the time of diagnosis and the start of HAART treatment, as CD4 counts can be affected by the treatment. Specifically, we estimate the following regression:

$$CD4_{it} = \phi_i + \beta_1 time + \beta_2 time^2 + \epsilon_{it}, \tag{1}$$

where *time* is years from diagnosis and ϕ_i is an individual fixed effect that captures differences in levels across the different individuals. The slope parameters β_1 and β_2 represent the annual changes in an individual's CD4 counts from the point of HIV diagnosis to the initiation of HAART treatment. We then use the estimated parameters to impute CD4 count values that are missing between the time of diagnosis and the start of HAART treatment, mainly pre-1995. We estimate the linear parameter β_1 to be equal to -31.5 with a standard error of 1.08 and the quadratic term β_2 to be equal to 0.57 with a standard error of 0.061. These estimates are in the lower end of the range of CD4 count declines for untreated individuals reported in the medical literature (Rodríguez et al. (2006), e.g., report rates of decline ranging between -80 and -20).[15] The high accuracy of our imputation method is illustrated in Appendix Figure A.1, which compares our predicted CD4 count values to observed values for the post-1995 period across CD4 count levels, time since diagnosis, and age.

3.3 The effect of HIV on health

This section explores our unrestricted sample, meaning the full population of HIV-positive individuals with all levels of CD4 counts, to document general patterns relating HIV progression to CD4 counts and health outcomes. These findings inform our definition of the final analysis sample, as described in the next subsection.

First, we illustrate the evolution of CD4 counts over time after HIV diagnosis for cohorts of individuals according to the year in which they were diagnosed. Appendix Figure A.2 clearly shows that CD4 counts decrease after diagnosis for individuals

[15] Although CD4 trajectories differ somewhat across individuals, the overall variation is limited. This is consistent with medical evidence indicating that CD4 progression is driven mainly by genetics and viral load, with observable characteristics (beyond age) contributing little to heterogeneity (Jia et al., 2020; Glaubius et al., 2021).

diagnosed in early years, before HAART medication was introduced, while the introduction of HAART in 1995 halts this reduction and even allows a recovery in CD4 count levels.

Second, we illustrate how the prevalence of health symptoms increases at low levels of CD4 counts, when HIV is at a more advanced stage. Specifically, we show in graph (a) of Figure 2 that the number of hospital visits (excluding those related to HIV) rises significantly once CD4 counts fall below 250, while remaining statistically indistinguishable from those of individuals without HIV at higher levels. Similarly, graph (b) shows a significant increase in the Charlson index for CD4 counts below 200, with a modestly higher prevalence for levels between 200 and 300.[16] Graph (c) also suggests a slight increase in infections for CD4 counts below 350, but less precisely estimated. Graph (d) shows imprecisely estimated results for an indicator of mental health, suggesting low levels of mental health care use and limited variation across CD4 count levels. Overall, these findings are consistent with the medical literature and guide our definition of the analysis sample as described next.

3.4 Sample selection

To define our analytical sample, we impose four restrictions starting from the unrestricted sample of all individuals diagnosed with HIV. First, we select the 1,943 individuals who were diagnosed with HIV either in the four years preceding the introduction of HAART (1990-1993) or in the five years following its introduction (1995-1999), who make up our control and treatment groups, respectively. Note that individuals in the control group are dropped from the analysis as they reach 1995 to avoid contamination, as they also become aware of HAART. For this reason, those diagnosed in 1994 are excluded from the main analysis. We demonstrate in the robustness section that our results are robust to the inclusion of those diagnosed in

[16] The Charlson Index is a weighted index that predicts mortality on the basis of pre-existing conditions. We follow Quan et al. (2011) in defining the index as a weighted sum of comorbidity conditions based on the ICD-10 classification. These include congestive heart failure, dementia, chronic pulmonary disease, connective tissue disease, liver disease, diabetes, hemiplegia, renal disease, tumors, leukemia, and lymphoma. We exclude HIV from the index.

1994 in the control group.

Second, we exclude individuals who are likely to be drug addicts according to their source of HIV infection, as we expect the behavior of these individuals to differ markedly from the rest of the sample. This leaves us with 1,743 individuals.

Third, we restrict the sample to those with a healthy immune system at the time of diagnosis, who are thus not expected to suffer from any HIV-related physical symptoms in the years following the diagnosis. Specifically, we keep individuals whose CD4 count levels are equal to or above 400. This reduces the sample to 536 individuals, 229 of whom were diagnosed before 1994 and 307 from 1995.

Fourth, we balance the sample by only including individuals who are observed annually from 4 years before diagnosis until 4 years after, resulting in a final sample of 402 individuals. The control group comprises 189 individuals diagnosed before 1994, while the treatment group comprises 213 individuals diagnosed from 1995.

3.5 Descriptive Statistics

Columns 1 and 2 of Table 1 show summary statistics for key background and outcome variables for our sample. The background variables are measured one year before the diagnosis, while the CD4 count is measured in the year of diagnosis. The treatment group comprises individuals diagnosed with HIV between 1995 and 1999, after HAART became available, while the control group includes those diagnosed between 1990 and 1993, before the advent of HAART. Some notable features of the sample are that males are heavily overrepresented (about 82 percent) and that the average age is about 34. Heterosexual individuals represent 42 percent of the sample.

Column 3 provides a comparison of means between the treatment and control groups, while column 4 reports the *p-values* for tests of equality of means. These tests indicate that the treatment and control groups exhibit substantial similarity across most observable characteristics, and a joint test for all variables reported in the table cannot reject the hypothesis of equal means with a p-value of 0.22. Among the 17 variables considered, only the indicator for marriage shows a difference that is significant at the 5% level, although we note that the evolution of this variable before diagnosis follows a parallel trend between treatment and control, as we show in the results section. The similarity between the treatment and control group is reassuring, as the existence of HAART treatment could affect risk behaviors, as suggested by Chan et al. (2015), and thereby could have changed the composition of HIV-positive individuals after 1996.

Finally, column 5 displays the means for the HIV-negative matched synthetic control group, which we further describe in Section 4. In terms of employment and earnings, the HIV-positive individuals have a pre-diagnosis employment rate of around 68 percent, which is lower than the 79 percent employment rate of the HIV-negative matched group. Earnings are also lower: HIV-positive individuals earn around 27,000 DKK less than their HIV-negative counterparts. Compared to the HIV-negative sample, HIV-positive individuals have a similar level of education, are less likely to own a house, less likely to be in a partnership, and display similar health markers as the HIV-negative matched sample.

3.6 Balancing on Health

A key feature of our empirical design is our focus on HIV-positive individuals who exhibit high CD4 counts, indicating good health and an absence of physical symptoms. By focusing on this sample, we can be certain that any sharp differences in life expectancy arising from differential access to HAART treatment do not coincide with differences in physical symptoms. To substantiate this claim, we present four lines of evidence.

First, we demonstrate that the CD4 count at the time of diagnosis of individuals in both the treatment and the control groups was remarkably similar. Table 1 shows that the average CD4 count at the time of diagnosis was 615 for the control group and 620 for the treatment group, well above thresholds where any HIV-related symptoms appear (see Figure 2). Furthermore, panel (b) of Figure 3 shows that average CD4 count at the time of diagnosis did not change over time, and panel (c) shows that the distribution of CD4 count at the time of diagnosis was very similar for both groups. Moreover, as depicted in Figure 4, a negligible share of individuals in our sample showcase CD4 counts below 200 in the four years post-diagnosis. Even when applying a more conservative threshold of 300, only a small fraction of individuals in both groups manifest CD4 counts below this threshold.

Second, Table 1 reveals that the treatment and control groups are statistically similar across various health indicators assessed prior to the HIV diagnosis. Both groups have comparable rates of infectious diseases and similar Charlson Index scores. Furthermore, the treatment and control groups exhibit comparable rates of hospital visits, excluding HIV-related visits, and these rates are similar to those observed in individuals without HIV. In the subsequent results section, we also present evidence indicating that there are no statistically significant differences in health outcomes between the treatment and control groups in the years following the HIV diagnosis.

Third, the treatment and control groups face high and comparable survival rates during the first years after diagnosis, as demonstrated in panel (d) of Figure 3. Note that, by construction, individuals in the control group must survive at least one year after diagnosis to be included in the medical database and in our sample. If we impose the same requirement on the treatment group, the survival curves align even more closely, as indicated by the dotted line.[17]

Fourth, while the appearance of HAART medication might have an effect on the incentives to get tested for HIV (Wilson, 2016), which could potentially affect the composition of our control and treatment groups, we do not find any significant differences along a range of observable characteristics reported in Table 1, such as age, gender, education, income, or home ownership, as well as along the medical variables already mentioned. Furthermore, we show in panel (a) of Figure 3 that rates of diagnosis remained stable during the period of analysis (1990-1999), with no observable increase in diagnosis after 1995. We further show in Appendix Figure A.3 that the take-up of HAART was remarkably similar for our treatment and control groups over calendar time, with the take-up starting in 1995 and growing rapidly for both groups afterward. HAART take-up was, of course, different between treatment and control groups when measured relative to the year of diagnosis due to its differential

[17] Note that in our baseline sample of analysis, we require all individuals to survive for at least four years after diagnosis.

availability. This differential take-up relative to diagnosis should not affect health, given the high CD4 of the analysis sample, but we show in the robustness section, nevertheless, that our results are robust to dropping individuals who ever receive HAART during the period of analysis.

3.7 Life Expectancy in the Treatment and Control Groups

For the validity of our empirical design, it is crucial that the treatment and control groups perceive significant differences in life expectancy. Although Figure 1 indicates substantial overall differences between those diagnosed before and after 1995, the difference between our treatment and control groups will be smaller. This is because our sample consists of healthy but HIV-infected individuals with an average CD4 count of 620, whereas Figure 1 includes *all* HIV-infected individuals.

While it is challenging to determine the exact life expectancy perceived by patients in our analysis sample, we estimate that individuals in the control group could expect to live around 16 years. This estimate is based on observed CD4 count progression from Equation (1), which suggests that without HAART, individuals' CD4 counts decline by approximately 30 each year. Therefore, for an individual with an initial CD4 count of 620, as in our sample of analysis at the time of diagnosis, they would see their count drop below 200 in approximately 14 years, at which point most patients die within two years.

Conversely, the life expectancy of the treated group was significantly improved as soon as the medication proved effective in increasing their CD4 levels. Indeed, we observe that more than 76 percent of individuals in the treatment group were alive 20 years after their diagnosis. Therefore, it is clear that life expectancy, and likely perceptions of it, varied dramatically between our treatment and control groups.

4 Methodology

Our empirical approach involves comparing the evolution of outcomes of individuals diagnosed with HIV before and after the introduction of HAART in 1995. Specifically, we consider individuals diagnosed between 1995 and 1999 as our treatment group, and those diagnosed between 1990 and 1993 as our control group. We exclude observations from the control group from 1995 onward, as they become aware of the introduction of HAART. For this reason, individuals diagnosed in 1994 are excluded from our baseline analysis, as their control status becomes contaminated just after diagnosis. To ensure that the groups are healthy and have not experienced immune system deterioration, as documented in Section 3.6, we restrict our sample to HIV-infected individuals with high CD4 counts.

Since the treatment and control groups are observed in different years, a simple comparison would risk confounding the effects of increased life expectancy with other factors that change over time. To control for such calendar time effects, we construct and match additional synthetic control groups of individuals not infected with HIV (HIV–). We construct these synthetic control groups by matching 1,000 HIV– individuals of the same cohort, age, and gender for each HIV+ individual in our sample. The matches are based on characteristics of the HIV+ individuals observed four years before diagnosis, and the matched HIV– individuals are then followed over time, preserving the panel structure of the data.

With our matched synthetic controls, we estimate the following standard dynamic triple difference specification:

$$\begin{aligned} Y_{it} =& \alpha_0 + \sum_{j\neq -1} \beta_j \cdot Treat_i \cdot Inf_i \cdot Time_{j=t} + \sum_{j\neq -1} \gamma_j \cdot Inf_i \cdot Time_{j=t} \\ &+ \sum_{j\neq -1} \eta_j \cdot Treat_i \cdot Time_{j=t} + \sum_{j\neq -1} \theta_j \cdot Time_{j=t} \\ &+ \phi_1 \cdot Treat_i \cdot Inf_i + \phi_2 \cdot Inf_i + \phi_3 \cdot Treat_i + X_{it} \cdot \Phi_4 + \epsilon_{it}, \end{aligned} \tag{2}$$

where Y_{it} is the outcome variable of interest for individual i in time t, $Treat_i$ is a dummy variable that takes value one if an individual is diagnosed with HIV in the period 1995–1999 when HAART was available, and zero if the individual is diagnosed with HIV in the period 1990–1993 when HAART was not yet available, $Time_{j=t}$ is a dummy variable equal to one if the year since the diagnosis is equal to t, and Inf_i is an indicator that takes one if an individual is ever infected with HIV and zero otherwise, that is if the individual belongs to the synthetic sample of individuals who are not diagnosed with HIV.[18] X contains the control variables: age dummies, gender, and a dummy for being a Danish citizen.

The β_j coefficients identify the causal effect of the introduction of HAART medication on various outcomes. By plotting β_j over time t we are able to evaluate the identifying assumption that both treatment and control groups move in parallel before the HIV diagnosis that occurs at $t = 0$. We present and discuss these graphical results in Section 5.

To quantify the average effect of the introduction of HAART, we also estimate a static version of the previous equation, which differs only in that the dummy variables for time since diagnosis, $Time_{j=t}$, are now replaced by a single dummy variable, $Post$, that takes the value one for all years after diagnosis, including $t = 0$ and up to $t = 3$.

$$\begin{aligned} Y_{it} =& \beta_0 + \beta_1 \cdot Treat_i \cdot Inf_i \cdot Post_{it} + \beta_2 \cdot Inf_i \cdot Post_{it} + \beta_3 \cdot Treat_i \cdot Post_{it} \\ &+ \beta_4 \cdot Post_{it} + \beta_5 \cdot Treat_i \cdot Inf_i + \beta_6 \cdot Inf_i + \beta_7 \cdot Treat_i + X_{it} \cdot \Gamma + \epsilon_{it}. \end{aligned} \tag{3}$$

In Section 5.5, we investigate the robustness of our main findings by addressing potential concerns related to our empirical strategy and sample definition, such as dropping individuals who ever receive HAART during the period of analysis, controlling for CD4 count and year of diagnosis, and keeping those diagnosed in 1994. Overall, our robustness checks provide strong support for our main findings, suggesting that our results are not driven by any particular specification or assumption.

5 Results

In this section, we report our main findings. To illustrate the dynamic effects of the treatment, we first present the results in four-field figures that show the contrasts

[18]Each individual of the synthetic sample of HIV-negative individuals is assigned the same value of $Treat_i$ as the HIV-positive individual to whom they were matched as well as a relative time to diagnosis t.

used in the Triple Difference specification. Additionally, we provide static regression results based on Equation (3).

For each of the main outcomes of interest, we report a four-field figure with the dynamic results. Specifically, graph (a) plots the evolution of the outcome before and after the HIV diagnosis for the control group (those diagnosed with HIV between 1990 and 1993, before the introduction of HAART) against the corresponding evolution for the matched synthetic control group of HIV-negative individuals. In Graph (b), we show the corresponding evolution for the treatment group (those diagnosed with HIV between 1995 and 1999, after the introduction of HAART) and its matched synthetic control of HIV-negative individuals. Graph (c) plots the evolution of the control and treatment groups, demeaned by their respective synthetic control groups. Finally, Graph (d) presents the event study estimates, β_t, of the triple difference model estimated in Equation (2), which identifies dynamic causal effects of the introduction of HAART treatment.

5.1 Labor Market Outcomes

We begin by studying whether having access to HAART around the time of the HIV diagnosis, and thus facing a higher life expectancy, affects labor market outcomes. Figure 5 displays the results for employment, where we observe that the control and treatment groups faced similar employment trends that were slightly declining in the years before diagnosis (graphs (a) and (b), respectively). We also observe that the matched synthetic control groups based on individuals not infected with HIV display slight differences in their trends, which are declining for those matched to the control group (corresponding to earlier calendar years) and flat or slightly upward sloping for those matched to the treatment group (later calendar years). This divergence highlights the importance of bringing in this additional control group to account for calendar time effects. At the time of diagnosis, then, the employment of HIV-infected individuals differs between the treatment and control groups, where the control group further reduces their employment while the treatment group barely sees any reduction. This difference in employment trends suggests that access to

HAART may have a positive impact on employment.

The divergence in employment trends between the treatment and control groups at the time of diagnosis becomes even clearer in Graph (c), where we plot the outcomes of the two groups, demeaned by their respective synthetic control group. The treatment and control groups closely follow each other until the year of the diagnosis, after which they start to diverge. Recall that the divergence cannot be explained by any divergence in health between the groups, as the analysis focuses on HIV-positive individuals who are still in good health, where the share of individuals with CD4 counts below critical thresholds for symptoms is similar and very low in both groups for the follow-up period considered, as shown in Figure 4. Rather, the divergence likely reflects the sharp differences in life expectancy between the groups, which may affect labor supply incentives.

Graph (d) presents the event study estimates from the triple difference specification and provides a similar picture. The estimates confirm that the groups face parallel employment trends before the diagnosis but depart afterward. The employment rate is about 10 percentage points higher in the treatment group than in the control group in the second year after diagnosis and onward. On average, in the years following the diagnosis, the treatment group experiences a 5.5 percentage point increase in employment, corresponding to an 8 percent difference (Panel A of Table 2).

We proceed to examine the impact on labor earnings, recognizing that a significant shift in life expectancy has the potential to influence decisions regarding labor supply, both at the extensive and intensive margins, while also impacting overall productivity. We begin with an unconditional measure of earnings, capturing responses along all margins, followed by a measure that conditions on employment.

Figure 6 shows substantial effects on unconditional labor earnings. Before the diagnosis, the treatment and control groups exhibit similar trends, closely aligning with the trends in the synthetic control groups (Graphs (a) and (b)). However, after the diagnosis, Graph (c) reveals a sharp divergence, with earnings in the control group declining sharply. The triple difference estimates in Graph (d) indicate substantial effects: earnings in the treatment group are up to 20,000 DKK higher in the year following the diagnosis, with no comparable difference observed in prior years. Table 2 further confirms this pattern, showing an average post-treatment effect of 11,413 DKK, equivalent to an 8 percent increase in earnings (Panel A). We note, however, that these estimates are imprecise and only significant at the 5% level in the year right after diagnosis.

Figure 7 focuses on individuals who remained active in the labor market throughout the study period. The estimated effect closely mirrors that of the unconditional earnings measure, with average earnings in the treatment group surpassing those in the control group by 16,059 DKK in the years following the diagnosis (Table 2). This difference is statistically significant at the 10 percent level and indicates that labor supply responses are not limited to the extensive margin.

Taken together, the earnings results suggest that the primary role of HAART is to prevent an earnings collapse at the time of diagnosis, rather than to generate substantial post-diagnosis gains above prior levels. In the control group, the sharp drop in earnings is consistent with a shorter expected lifespan, making future income less valuable — both because there are fewer years left to spend it and because the returns to career investment shrink, which increases the relative attractiveness of leisure and encourages early withdrawal from the labor market. For those with access to HAART, the preserved value of future earnings gives individuals a strong reason to remain engaged in work and maintain their pre-diagnosis earnings levels.

An alternative explanation for the observed differences could relate to workplace discrimination. Before HAART, fears of infection may have led some employers to discriminate, whereas the introduction of HAART greatly reduced the risk of transmission and thereby diminished such concerns. However, disclosure rates of HIV status to employers are generally low (see, e.g., Maulsby et al., 2020), and there has never been a legal obligation in Denmark. Recall also that our study population had not yet experienced health effects from the HIV infection, which makes it difficult for employers to infer individuals' HIV status. We therefore expect employer responses to play a limited role in explaining the earnings effects we document.

5.2 Savings, Housing, and Stock-holding

As the introduction of HAART dramatically changed the life expectancy of HIV-positive individuals, it also affected their financial investment horizon. We continue by investigating how the increased life expectancy affected a) bank account savings, b) stock-holding, and c) home ownership. Figure 8 shows the triple difference event study estimates for these outcomes.

Our results reveal that the group gaining access to HAART experienced a larger decrease in bank account savings (Graphs (a) and (b) in Figure 8). Table 2 shows that the effect amounts to a reduction of 12,495 DKK, corresponding to a 44 percent decrease. While this is a large effect, it is also imprecisely estimated and not significant. The graphs further illustrates this finding, as it shows that the effect is driven by a decline in the treatment group, while the savings in the control group appear to be unaffected by receiving an HIV diagnosis.

Did the increase in life expectancy encourage the treatment group to invest more in risky assets? Graphs (c) and (d) provide no strong evidence for such an effect, as the triple difference estimates show that stock ownership does not evolve very differently in the treatment and control groups. Table 2 reveals, however, that there was a 42 percent decline in stock ownership during the post-treatment years, but this change was statistically insignificant.

Did the change in life expectancy resulting from HAART treatment impact other significant long-term decisions, such as home ownership? We obtain no such evidence, as illustrated in graphs (e) and (f). The treatment and control groups follow similar trends, both before and after receiving an HIV diagnosis. The absence of an effect is confirmed in Table 2, where the average effect is insignificant.

Overall, the results suggest that the dramatically increased life expectancy due to HAART had limited effects on important economic decisions such as bank account savings, home ownership, and stock-holding. What can explain these results? While we acknowledge that the absence of significant effects may, in part, be due to limited statistical precision, there are also plausible substantive explanations for the lack of observed impact. First, recall that we study a relatively healthy group of

HIV-positive individuals, where death is not imminent, and who face uncertain life expectancies. While life expectancy after an HIV diagnosis is, on average, shorter in the control group, some individuals will live longer than expected. The fact that assets are not depleted in the control group could be attributed to precautionary saving motives, reflecting such uncertainty in remaining life expectancy. It is also plausible that the depletion of assets mainly occurs closer to death, when health has deteriorated. Figure A.4 in the appendix does not suggest so, however, as the patterns closer to death are similar in the treatment and control groups. Another explanation for the lack of asset depletion could be that individuals in our sample possess strong bequest motives. Furthermore, the absence of any effects could be attributed to the generosity of the Danish public pension system, where there is limited necessity to save for one's own retirement.

Taken together, the results from the analyses of earnings and the various wealth components suggest that consumption patterns may have been affected by the arrival of HAART. Notably, the sharp decline in earnings in the control group at the point of HIV diagnosis, while other significant wealth components remained unaffected, suggests that consumption levels were reduced. This could reflect a substitution from labor to leisure when otherwise healthy individuals in the control group learn about their limited life expectancy.[19]

5.3 Marriage market outcomes

Significant shifts in life expectancy can alter incentives for partnership formation. In Figure 9, we illustrate the effects on partnership formation, defined as marriage or cohabitation. In Graph (a), we observe a notable increase in the likelihood of having a partner in the control group immediately following an HIV diagnosis, whereas no such increase is observed in the synthetic control group. In contrast, in the treatment group, we observe a decrease in the likelihood of having a partner at the time of diagnosis (Graph b). The different patterns in the treatment and control groups

[19] Since we do not observe all components of wealth, we cannot rule out that some other components of wealth were depleted, in order to maintain the same level of consumption.

are further illustrated in Graph (c) and Graph (d), which present the triple difference estimates. These estimates demonstrate a negative effect of the increased life expectancy resulting from HAART on partnership formation, reflecting the differential patterns observed in the treatment and control groups. Table 3 shows that this negative effect amounts to about 62 percent across the post-treatment years.

How can we interpret the divergent patterns observed in the treatment and control groups? One possibility is to interpret the increase in partnership formation in the control group, who faced a negative shock to life expectancy following their HIV diagnosis, through the lens of family economics. According to this perspective, cohabitation and marriage function as a crucial form of intra-marriage insurance against health shocks (Anderberg, 2007; Persson, 2020; Potoms and Rosenberg, 2021). In this framework, as health deteriorates and the demand for care rises, having a partner becomes increasingly valuable for support and assistance. Another possible interpretation for the observed increase in partnership formation in the control group is that individuals who are HIV-positive and have few years left may prioritize leisure over consumption, thus increasing the utility of being in a couple. This interpretation is consistent with the results of the previous section, which showed that an HIV diagnosis was associated with decreased consumption in the control group. In contrast, individuals diagnosed after 1995 did not have an insurance motive for partnership formation, and their "option value of waiting" for the right match was higher due to the introduction of HAART and the resulting higher life expectancy. The decline in marriage and cohabitation in this group may also reflect social stigma, reduced attractiveness on the marriage market, and marriage turmoil following from HIV diagnosis.[20]

At the same time, an HIV diagnosis likely reduces an individual's attractiveness on the marriage market, as potential partners may be reluctant to enter a relationship with someone who is HIV positive. This demand-side constraint suggests that the observed increase in partnership formation among individuals in the control group may

[20]HAART also reduced the risk of spreading HIV infection through sexual activity. Some evidence suggests that the introduction of HAART may have led to an increase in the number of sexual partners, potentially contributing to marriage turmoil (Lakdawalla et al., 2006).

come at the expense of match quality. To examine this, we compare the income and education of individuals and their partners for those who formed a partnership after the diagnosis. Specifically, we calculate the relative difference in income and education (in months of schooling) between each individual and their partner—measured prior to diagnosis—and compare these differences across the treatment and control groups. We find that, after diagnosis, individuals in the control group tend to partner with spouses who have lower income (about 22 percent less) and around the same years of education (4 percent more). However, these estimates are based on a small subsample and are imprecise, so they should be interpreted with caution.

To further improve the understanding of the mechanisms behind the effects, it can be informative to study whether the effect on partnership formation reflects changes in marriage, cohabitation, or divorce rates. Table 3 reveals that both marriage and cohabitation rates are reduced in the treatment group. While it is possible that part of this decrease reflects an increase in divorces, the results shown in the last row of the Table rule out this possibility, as we estimate a very small and insignificant effect of 0.00236, indicating that divorces from marriage turmoil are not the driving factor.[21]

The observed increase in both marriage and cohabitation rates in the control group suggests that bequest motives are unlikely to be the main driving force behind the observed increase in partnership formation. Bequest motives would typically have a stronger effect on marriages, as the surviving spouse in Denmark automatically inherits the estate of the deceased spouse. To further explore this, we also examined whether the effects differ by sexual orientation, since the strength of any bequest motives may vary due to differences in the presence of children in the household. However, as demonstrated in the second and third columns of Table 3, the effects are significant and similar in size for both heterosexual and homosexual couples.

[21]We define divorce as a flow variable that takes the value one if an individual transitions from being married to a specific partner to being either non-partnered or married to a different person.

5.4 Alternative mechanisms

We continue by providing a set of results that shed additional light on potential alternative mechanisms behind the effects we estimate. First, we consider whether the changes in life expectancy had effects on mental and physical health. Second, we consider whether the increased life expectancy incentivized some individuals in the treatment group to obtain additional education, as the returns to education increased.

5.4.1 Impact of Mental and Physical Health on Behavior

In Section 3.4, we have shown that our treatment and control groups consist of HIV-positive individuals who were in good physical health, based on their CD4 counts at the time of diagnosis. However, one remaining concern is that drastic changes in life expectancy can have effects on mental health and mood that can potentially affect behavior. Specifically, a positive shock to life expectancy may improve mental health. While such changes in mental health can be part of the causal chain through which life expectancy affects behavior, it is a distinct mechanism from the pure incentive effect that arises from changes in life expectancy (Oster et al., 2013b). Furthermore, even though we have selected HIV-positive individuals who were in good health according to CD4 counts, it is important that the drastically increased life expectancy did not coincide with other type of changes in physical health.

Table 4 summarizes the findings from applying Equation (2) to both mental and physical health outcomes. Physical health is assessed using information on hospital visits, the Charlson Index, and infection rates. The results in Panel A indicate small effects on physical health outcomes, with none of the effects reaching statistical significance.

Mental health is assessed through visits to psychologists and psychiatrists at private clinics (with full or partial public subsidies). The estimates in Panel B, however, are small and statistically insignificant. It is important to acknowledge that our measures do not capture all dimensions of mental health, potentially overlooking changes that do not necessitate specialist care. Specifically, our metrics focus on severe mental health issues that require visits to mental health professionals, while lacking data on antidepressant use or on inpatient and outpatient hospital care in psychiatric wards.

Our results for mental health align with several studies from Western contexts showing that improvements in life expectancy due to medical innovations do not necessarily lead to measurable improvements in mental health. These studies have used a range of different measures of depression and psychological well-being, yet consistently find limited or no effects. Oster et al. (2013b) examined individuals at risk for Huntington's disease and found no significant difference in mental health outcomes between those who learned their status early versus late in life. Similarly, Papageorge (2016) studied the introduction of HAART in the U.S. and hypothesized that improved life expectancy might lead to better mental health among HIV-positive women. However, their results provided little evidence of changes in depression levels after HAART was introduced, and their main findings remained unchanged when controlling for mental health. Karparti (2022) found that individuals testing positive for a Lynch Syndrome mutation (a hereditary cancer risk) were actually less likely to consume antidepressants and anti-anxiety medication following genetic testing, suggesting that increased certainty—whether positive or negative—does not necessarily worsen mental well-being. These findings reinforce our interpretation that incentives are key drivers behind the behavioral responses to increased life expectancy, rather than short-term emotional reactions.

In contrast, Baranov et al. (2015) found that in Malawi, the expansion of HAART access did improve mental health outcomes among HIV-negative individuals, likely because it reduced uncertainty about AIDS mortality. Crucially, that study also showed that individuals updated their subjective life expectancy in response to the introduction of HAART, confirming that perceptions about survival probabilities—not just emotional relief—shape behavioral adjustments.

Taken together, these studies suggest that the incentive effects we document are unlikely to be driven solely by emotional responses to diagnosis. Instead, the evidence points to a more fundamental mechanism: when individuals anticipate living longer, they adjust their economic decisions accordingly, even in the absence of significant mental health changes. While we cannot entirely rule out that some behavioral adjustments reflect emotional improvements, the weight of the evidence—including findings from multiple Western studies—suggests that changes in economic incentives are the predominant driver of our results.

5.4.2 Education

The availability of a new health technology such as HAART may, by substantially extending life expectancy, increase incentives for individuals to invest in human capital, as shown by Oster et al. (2013a,b) in the case of Huntington disease. Human capital could, in turn, mediate the effects of longer life expectancy on outcomes such as marriage and savings. Although most individuals in our sample are beyond the age of pursuing substantial formal education, they could still respond through on-the-job training or shorter educational programs. We cannot directly observe such investments, but the positive effects on employment and earnings documented in Section 5.1 provide little support for this channel, since human capital accumulation would likely involve temporary earnings losses. Consistent with this, we also find no effects on years of schooling (Table 5). More broadly, health itself can be viewed as a form of human capital that shapes economic decisions (Grossman, 1972). Longer life expectancy should, in principle, strengthen incentives to invest in health, since the returns accrue over a longer horizon. Yet, as shown in Section 5.4.1, we find no significant improvements in health outcomes.

Taken together, these findings suggest that the behavioral changes we document are not driven by contemporaneous health improvements or additional human capital investments. This contrasts with both theoretical predictions and earlier empirical evidence, which highlight human capital accumulation as a central response to longer life expectancy (e.g., Oster et al. 2013b). The difference likely reflects the distinct age composition of our sample: unlike younger populations examined in prior work, our individuals are largely beyond the stage of making substantial educational investments. This makes our setting different, showing that when life expectancy rises later in life, behavioral adjustments occur primarily through labor supply, partnership, and savings decisions rather than through new human capital formation. More broadly, this distinction suggests that the role of human capital in mediating longevity effects is strongly age-dependent, with education shaping responses for the young, and altered economic incentives driving adjustments among older adults.

5.5 Robustness

We next demonstrate the robustness of our main results to alternative specifications and sample definitions. Specifically, for each of our primary outcomes (labor market, savings, marital, and health), we report in row A of Table 6 our baseline for comparison and in rows B to E alternative robustness exercises.

In row B, we add CD4 counts as control, and in row C, we add year of diagnosis as control. Note that these two variables do not exist for HIV-negative individuals since they are not diagnosed and do not appear in the DANHIV sample. Hence, we assume for them the same values as their matched HIV+ individual. We observe that the results are virtually unchanged compared to our baseline estimates.

Row D reports the results when individuals diagnosed in 1994 are kept in the analysis as part of the control group. Note that we exclude them from the baseline analysis because, since we exclude all observations from the control group from 1995 onward as they gain knowledge of the HAART medication, those diagnosed in 1994 would not be observed in any post-diagnosed year. We observe this makes little difference, with small reductions in some estimates but with unchanged significance levels.

Finally, row E shows the result of excluding individuals who have ever received HAART medication during the period of analysis. Individuals in the treatment group are more likely to receive HAART medication and receive it earlier relative to when they were diagnosed, in comparison with the control group. We show that our results are generally robust to this change, although the estimate for earnings conditional on staying employed becomes insignificant, while the (unconditional) estimate for earnings becomes significant.

5.6 Heterogeneity

We examine heterogeneity by age, gender, and education to better understand the mechanisms behind our main results. Differences across these groups can reveal how time horizons, opportunity costs, and access to resources shape behavioral adjustments to a life expectancy shock. All results are reported in Table 7.

5.6.1 Age

A sudden reduction in life expectancy can reduce labor market incentives for both younger and older individuals: the former lose many potential working years, lowering the payoff to long-term career investment, while for the latter, the return on continued work is small when only a few years remain, making earlier retirement or reduced effort attractive. In our setting, the interaction between treatment and age is small and statistically insignificant, indicating similar responses across age groups despite different horizons and adjustment possibilities.

From a life-cycle perspective, younger individuals might be expected to react more strongly in financial and partnership domains, while older individuals might spend down savings and place greater value on partnerships for care. Yet, opposing forces—such as similar insurance benefits of a partner for both groups, survival uncertainty, bequest motives, generous public support, and illiquid assets—can yield similar net responses. Consistent with this, we find no significant age differences in the effects on savings, financial assets, marriage, or cohabitation (Panel A).

5.6.2 Gender

Women, who on average have lower earnings than men, might be expected to reduce their labor supply more after diagnosis, as their lower opportunity cost of leaving work and greater caregiving responsibilities raise the relative value of leisure. However, countervailing factors—such as stable, lower-paid public sector jobs with rigid wage structures, existing part-time work, and pre-existing caregiving roles—limit scope for further reduction. In line with this, earnings declines are concentrated among men, though gender differences are not statistically significant.

In the control group, women's lower earnings heighten the need for precautionary saving and asset accumulation, as well as the value of self-insuring through a partner's income, assets, and caregiving. These motives are reinforced by higher rates of childbearing, which strengthen bequest motives, and by existing caregiving roles, which increase the value of reciprocal care. While the interaction terms generally suggest larger adjustments in savings, assets, and partnership among women, these differences are not statistically significant (Panel B).

5.6.3 Education

Low-educated individuals may respond more strongly in earnings because their jobs are often more physically demanding, less stable, and offer fewer long-term rewards. With a shortened horizon, continued effort in such jobs yields low returns, adaptation options are limited, and the financial cost of leaving is lower due to both lower earnings and higher social security replacement rates. In contrast, high-educated individuals have more adaptable jobs and stronger incentives to remain employed. Point estimates align with this story, but the differences are not statistically significant (Panel C).

For financial assets, no systematic differences emerge, with interaction terms generally insignificant. This may reflect offsetting forces: low-educated individuals may have stronger precautionary saving and bequest motives but also face tighter credit constraints and lower initial wealth, while high-educated individuals may preserve assets thanks to larger stocks yet also draw them down to finance near-term consumption or care.

In the partnership domain, low-educated individuals may have stronger incentives to marry or cohabit, as a partner provides valuable insurance against income and health shocks when self-insurance capacity is limited. High-educated individuals can more often self-insure, reducing the urgency of partnering. Here, interaction terms are statistically significant, consistent with stronger partnership responses among the low-educated.

Overall, the heterogeneity analysis suggests that while theoretical considerations often predict larger responses for certain groups—such as younger individuals, women, or the low-educated—many of these expected differences are muted in practice. In several cases, countervailing forces may offset each other, leading to similar net responses. Significant differences emerge mainly in partnership formation by education, consistent with the role of economic security and self-insurance capacity in shaping responses to shortened life expectancy.

5.7 The incentive effect and cost-effectiveness of new medical innovations

Our findings suggest that economic behaviors could be influenced by the anticipation of future access to life-extending medical technologies. Such incentive effects, which can manifest even before the onset of any health symptoms, are typically not accounted for in the assessment of the value of new medical technology (e.g., Davies et al., 1999; National Institute for Health and Care Excellence, 2025). In this section, we demonstrate how the benefits of new medical innovations may, therefore, be underestimated in general, using the expected employment gains from HAART as an illustrative example.[22]

To illustrate the incentive effect, we consider two hypothetical scenarios for HIV patients: one in which HAART is not yet available, and another in which HAART exists and treatment is initiated when the patient's CD4 counts fall below 350.[23] We assume that the patient in both scenarios is a Danish man, 34 years of age, with 11.9 years of education, and has a CD4 count of 500 at the time of HIV diagnosis. First, we estimate the development of CD4 over time with and without the HAART treatment. We then estimate the impact of CD4 counts on employment in each of the two scenarios, using data from the period 1995-2005. The details of the calculations are presented in Appendix C.

The top panel of Figure 10 displays the predicted development of CD4 in the

[22]The HAART treatment might have large incentive effects on labor supply because it induced a large increase in life expectancy for young patients. Medical innovations leading to small increases in life expectancy could have a limited incentive effect if it is costly to adjust labor supply.

[23]This was the guideline in the 1990s.

two scenarios. In the absence of HAART, CD4 counts decline steadily over time. With access to HAART, CD4 counts decline for the first five years following the HIV diagnosis until they reach 350, after which HAART treatment is initiated, and CD4 counts start to increase again. Using these predicted CD4 counts, we can calculate the implied employment for both scenarios. The lower panel of Figure 10 shows that even in the period before HAART treatment is initiated (when the CD4 counts are similar in both scenarios), the employment level in the scenario with HAART is higher. This difference is due to the incentive effect of higher life expectancy, resulting from individuals anticipating future access to HAART at the time when they will experience HIV symptoms (dark gray area). The incentive effect is estimated to be about 0.49 years of employment during the first five years.

During the period when HAART has been initiated, the difference between the two scenarios is due to both a pure health effect and the incentive effect (light gray area). The accumulated difference in employment between those who receive HAART and those who do not, from 5 to 15 years after the HIV diagnosis, is 1.83 years of employment. The total employment benefit of HAART treatment is therefore the sum of the health and incentive effects, but only the former is typically accounted for. In this example, the total difference in employment between those with access to HAART and those without is 2.33 years of employment during the first 15 years after diagnosis (the light grey and dark grey areas). This calculation indicates that the total employment effect of HAART treatment during the first 15 years is underestimated by about $0.49/(1.83+0.49) = 21$ percent if the incentive effect that is present during the period before HAART is initiated is not accounted for. This back-of-the-envelope calculation demonstrates that the full benefits of introducing medical technologies, such as HAART, may be underestimated unless both types of effects are accounted for.

6 Conclusion

Over the last century, populations in the Western world have experienced remarkable gains in life expectancy (Case and Deaton, 2020). This increase in life expectancy alters individual incentives to work, save, and marry, but it has proven challenging to distinguish the pure incentive effect of a longer planning horizon from that of improved health, as the two usually go hand in hand. We overcome this challenge by examining the impact of a sudden and dramatic rise in life expectancy caused by a crucial medical innovation: HAART treatment for HIV. By focusing on a sample of HIV-positive individuals who were still in good health but who faced different access to HAART, we can observe how otherwise healthy individuals react to sharp differences in life expectancy, allowing us to isolate the pure incentive effect of increased life expectancy.

Our empirical results reveal that the rise in life expectancy that followed the introduction of HAART had substantial effects on the labor market behaviors of HIV-positive but otherwise healthy individuals. Labor supply and earnings were considerably higher for those diagnosed after HAART became available, suggesting that the increase in life expectancy strengthened incentives to work. HIV-positive individuals without access to HAART, however, may have substituted labor for leisure, given their short remaining life expectancy.

On the other hand, we find only limited evidence that the positive shock to life expectancy had any large effect on financial decisions, such as savings and stock market participation among HIV-positive individuals. While perhaps surprising in light of standard life-cycle theory, we attribute this finding to strong bequest and precautionary savings motives.

Our results also indicate that the introduction of HAART had an effect on marriage market incentives. Partnership formation after an HIV diagnosis declined for those with access to HAART, while it increased for those without access. We interpret this finding through a family economics lens, where cohabitation and marriage serve as vital sources of private insurance against health shocks.

It is important to recognize that while our study highlights the impact of increased life expectancy on labor, savings, and marriage decisions, the significant improvements in life expectancy due to HAART may also have influenced mental health, potentially affecting economic behaviors. Although our findings are consistent with those of Oster et al. (2013b) and Karparti (2022), showing limited evidence that mental health plays a crucial role in the relationship between increased life expectancy and economic behavior, it is important to note that our measures do not encompass all dimensions of mental health. Consequently, our results may reflect both incentive effects and some influence of improved mental health.

Our research adds to the literature showing how longer life expectancy shapes individuals' incentives and economic behavior. While we do not find changes in formal schooling—consistent with education no longer being a relevant margin in this setting—we show that increased longevity affects labor-market and partnership decisions. Our results therefore complement the earlier human-capital–focused literature by demonstrating that life expectancy influences welfare and economic growth even in contexts where observable schooling investments do not respond.

Our results also speak to a broader insight: economic behavior is often shaped by expectations about future health and longevity, not just by present conditions. Even in the absence of symptoms, individuals may adjust their work, savings, or partnership decisions based on whether they believe effective treatment will be available. This mechanism is likely to be relevant well beyond HIV, particularly in cases where access to medical innovations is uncertain due to financial constraints, eligibility rules, or insurance coverage. While our study focuses on relatively young individuals facing a chronic illness, similar forward-looking responses may be observed among older populations, for example, in retirement planning or long-term care decisions. Recognizing these incentive effects is crucial when evaluating the full economic impact of life-extending medical technologies.

Finally, our findings have important implications for the valuation of life-extending medical technologies. Cost-effectiveness analyses typically do not account for the incentive effects that arise from increases in life expectancy due to new medical technologies, such as HAART. Based on our estimates, we conduct a simple back-of-the-envelope exercise, showing that the incentive effect can account for as much as 21 percent of the total effect on employment during the first 15 years after an HIV diagnosis. This highlights the importance of considering labor market incentives when evaluating the value of life-extending medical technologies, as failure to account for these effects may lead to underestimation of their true value.

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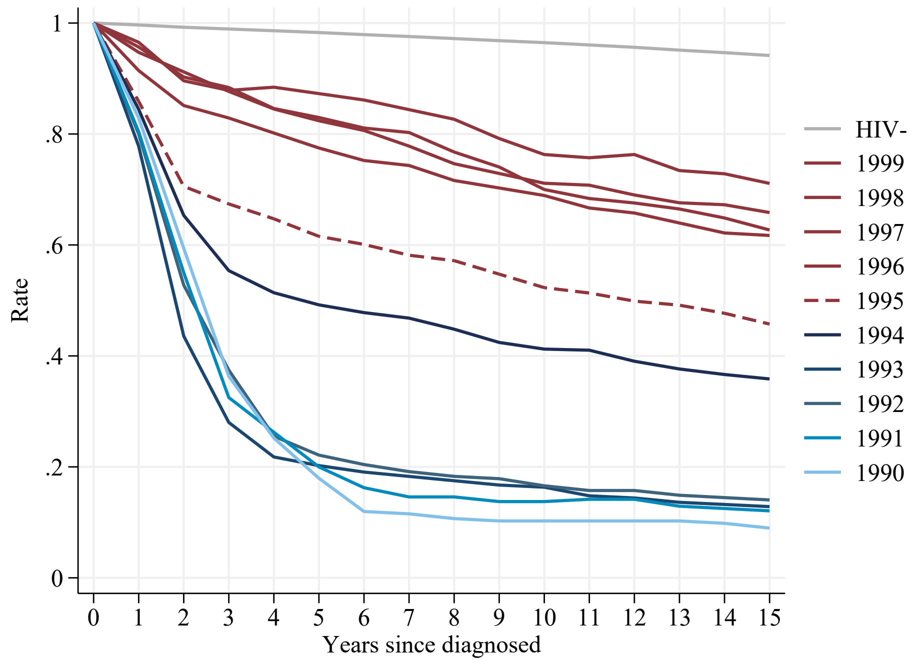
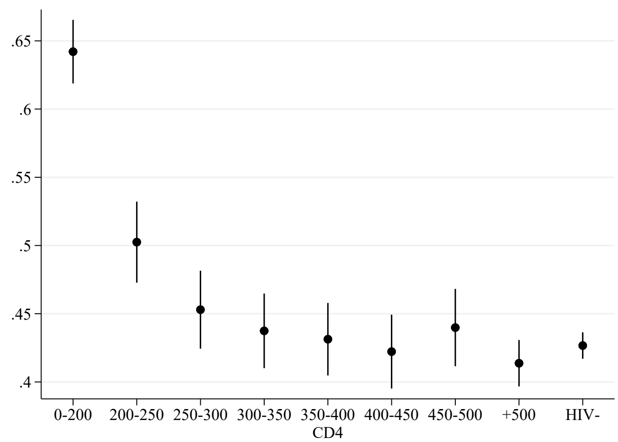
Figure 1: Survival by year of HIV Diagnosis, unrestricted sample



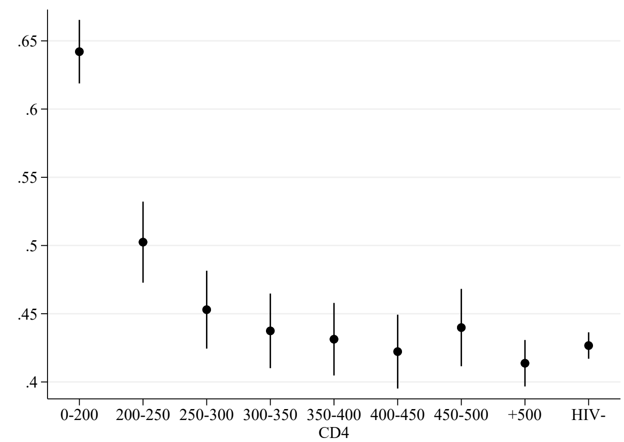
Notes: This figure plots the survival rate of all individuals diagnosed with HIV, distinguishing by year of diagnosis. The Figure illustrates that individuals diagnosed during earlier calendar years face sharp drops in their survival rates in the years following the diagnosis, while individuals diagnosed later, after the introduction of HAART medical innovation in 1995, face much improved survival rates. The gray line plots, for reference, the survival rates of a sample of individuals not diagnosed with HIV. The Figure is constructed using Danish hospital records on all HIV diagnoses (*landspatientregisteret*), which is not affected by any break or change of definitions during the period considered.

Figure 2: Health and HIV progression, unrestricted sample

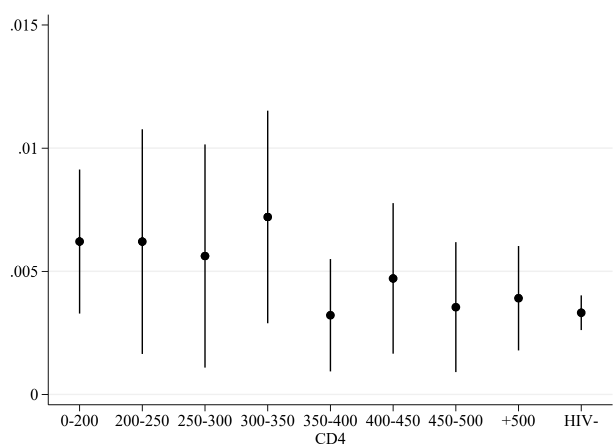
(a) Hospital visit



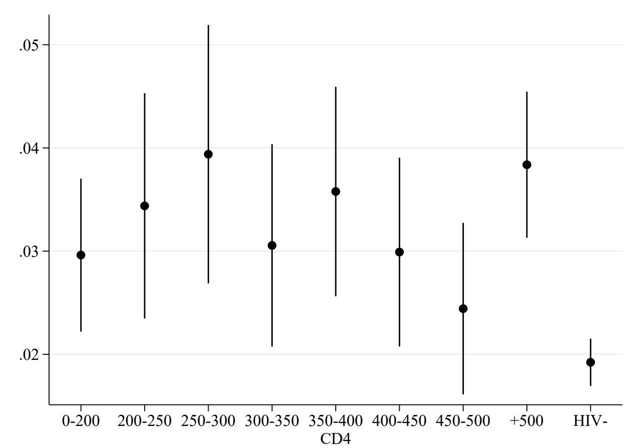
(b) Charlson Index



(c) Infections

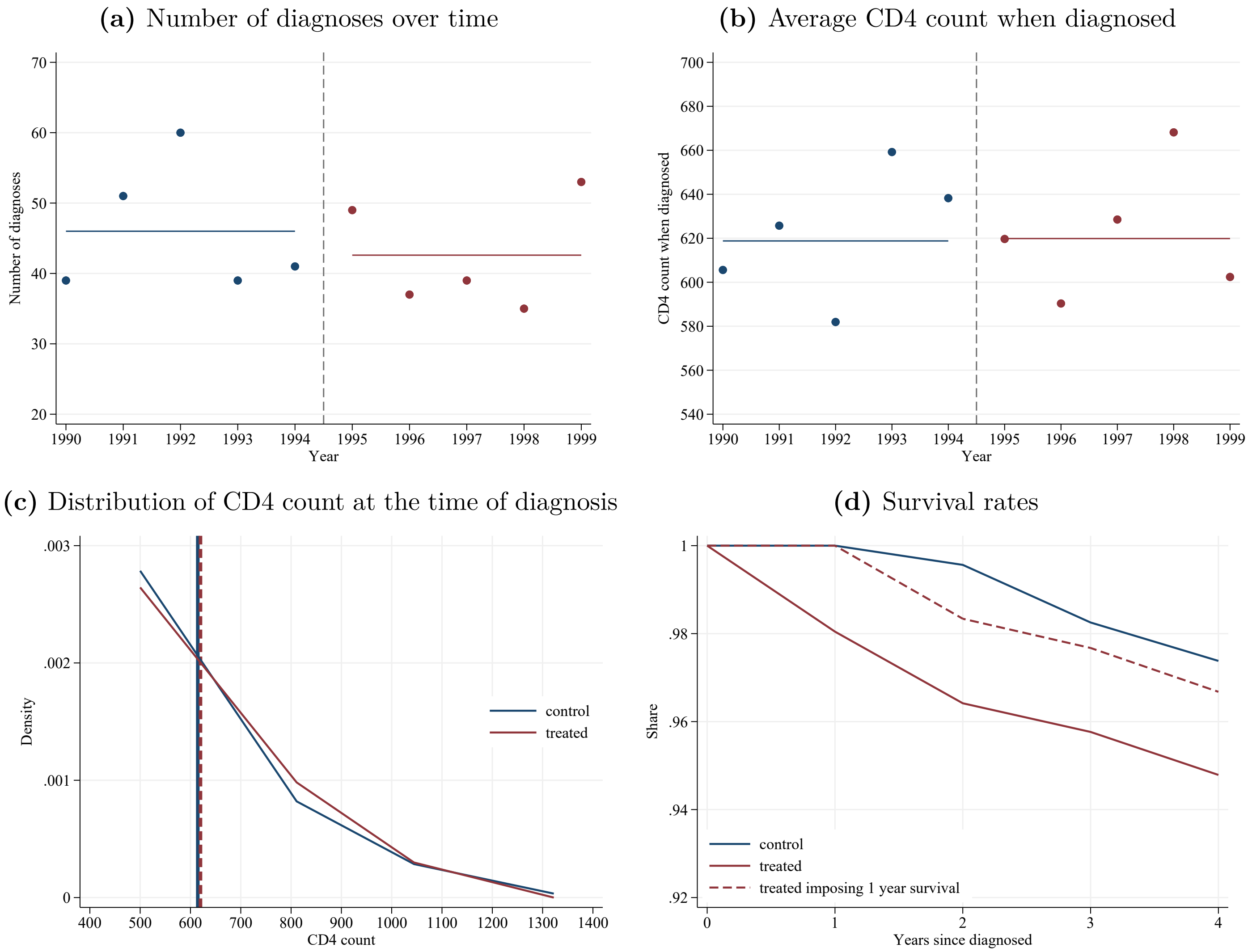


(d) Psychologist/psychiatrist



Notes: These graphs shows the relation between CD4 counts and different health outcomes: rate of hospital visits, Charlson index, infections, and mental health (see the definition of these outcomes in Section 3). The graphs illustrate how health outcomes change with the level of CD4. The estimates are obtained from the estimation of the health outcome on dummies of CD4 count groups and a dummy for HIV-negative (HIV–) while controlling for age, gender, ethnicity, education, and calendar year dummies. The estimation is based on the unrestricted sample (see Appendix C, Table C.1).

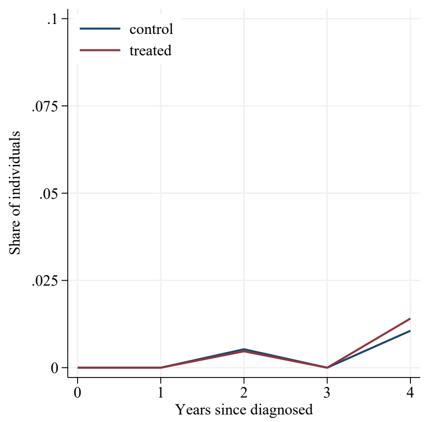
Figure 3: Selection and balancing of the analysis sample



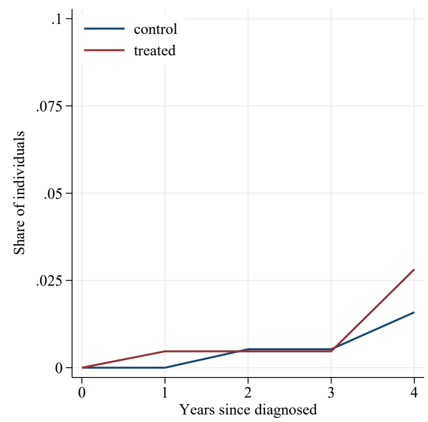
Notes: These graphs plot different statistics of interest related to the sample of analysis (individuals diagnosed with HIV when their CD4 count was at least 400 and who are observed at least during the 4 years before and after diagnosis) distinguishing by individuals in the control group (those diagnosed between 1990 and 1993) and in the treatment group (those diagnosed between 1995 and 1999). Year 1994 is shown for completeness. Panel (a) shows the number of diagnoses per year, represented by each dot, as well as the average for the two periods, represented by the solid horizontal lines. Panel (b) shows the average CD4 count at the time of diagnosis per year of diagnosis. Panel (c) shows the distribution of CD4 count levels at the time of diagnosis for the treatment and control groups. The vertical lines indicate the average CD4 count for each of the two groups. Panel (d) shows the survival rates of individuals in the treatment and control groups without imposing the balancing restrictions that individuals must be observed 4 years before and after their HIV diagnosis. Note that by construction, individuals in the control group must survive for at least one year (until 1995) to be included in the analysis. The dashed lines illustrate that imposing the same restriction on the treatment group leads to very similar survival probabilities, which are high because we restrict our analysis sample to individuals with CD4 counts above 400 at the time of diagnosis.

Figure 4: Share of Treated and Control Individuals Below CD4 Count Thresholds

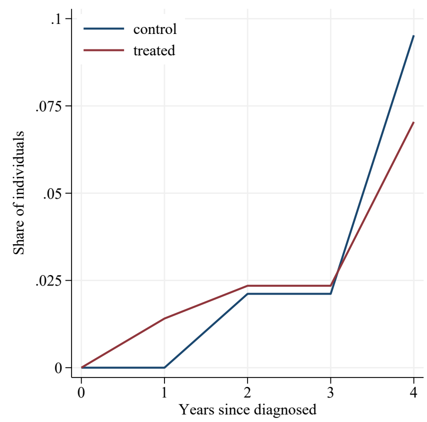
(a) Under 200



(b) Under 250

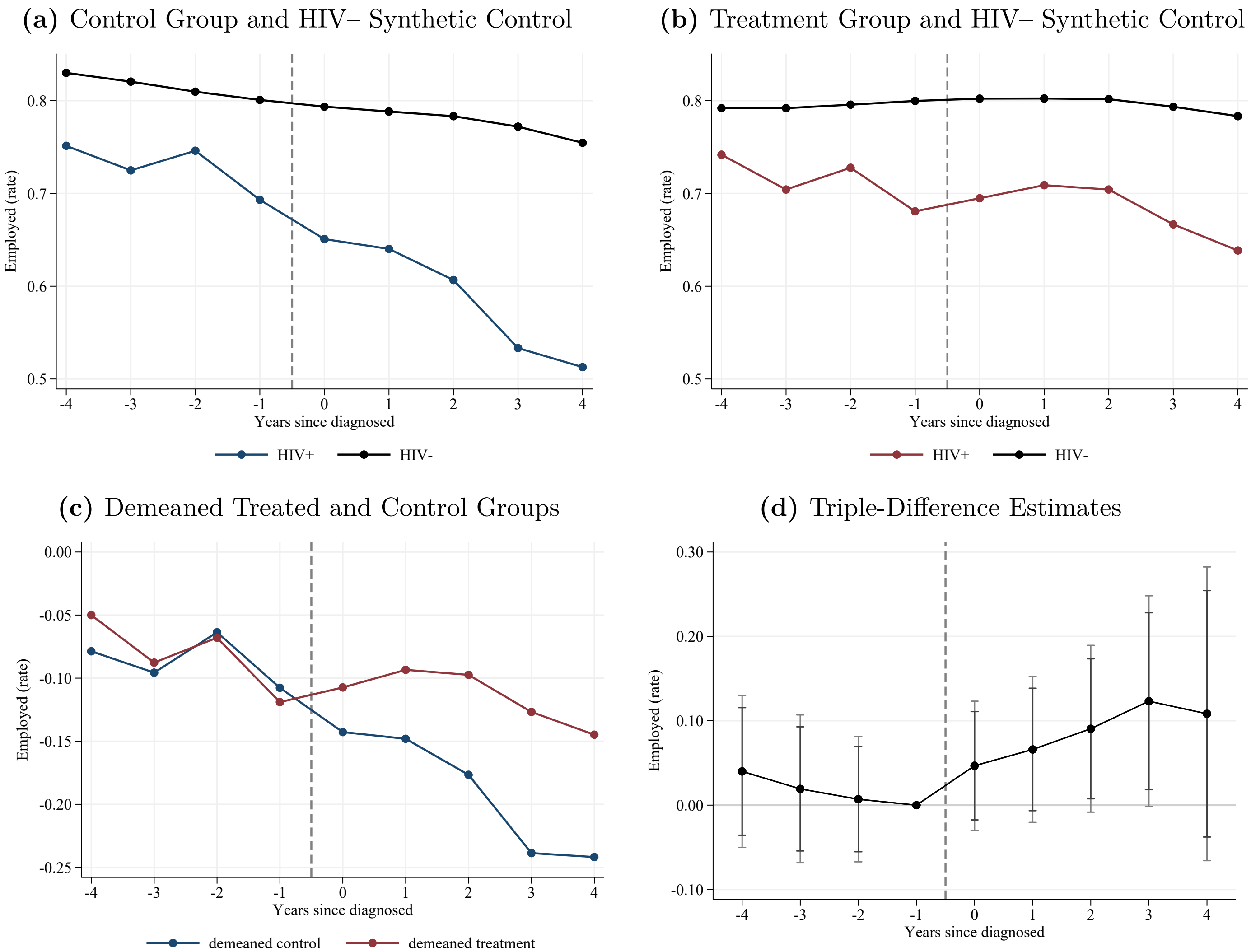


(c) Under 300



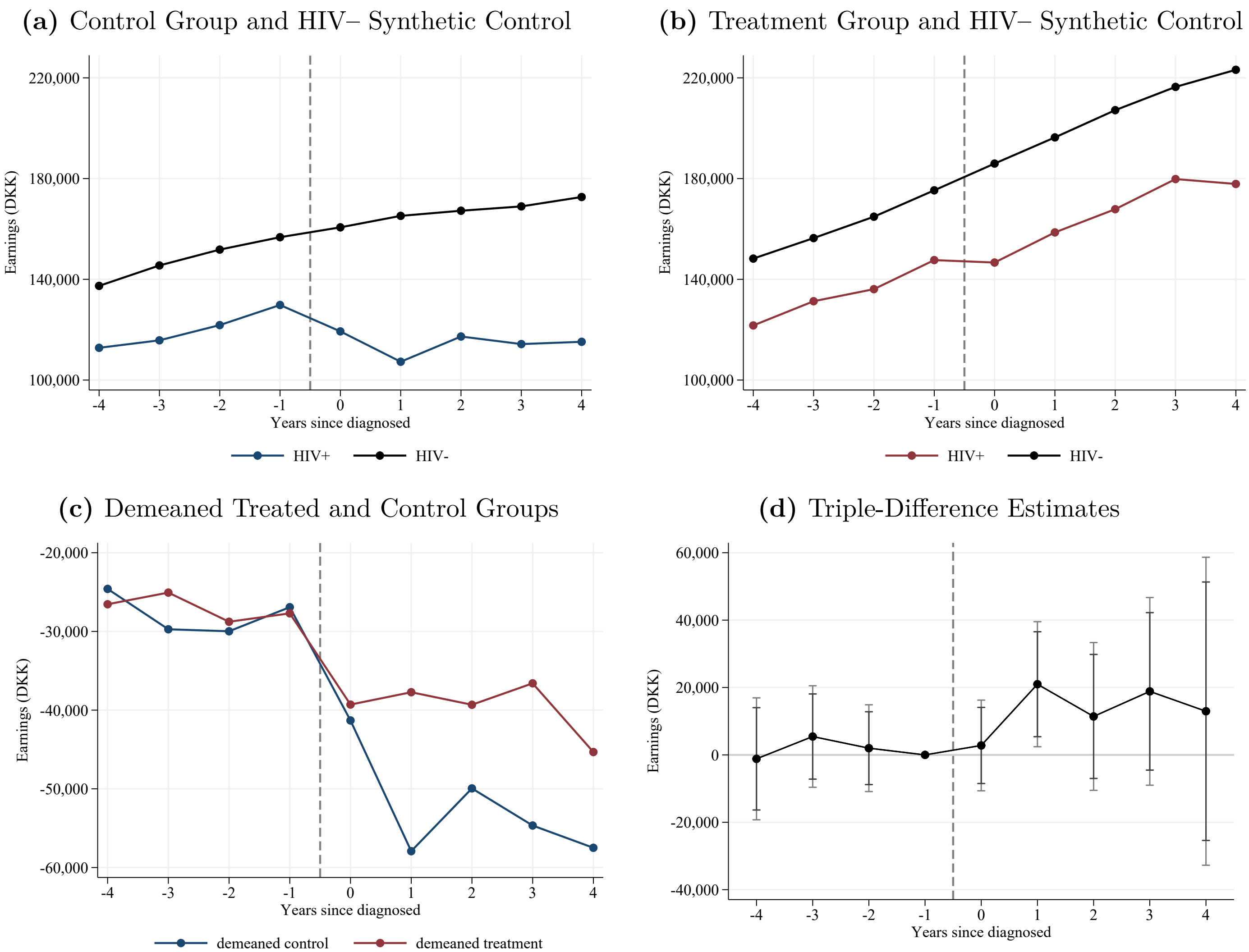
Notes: These figures display the proportion of individuals in the treatment and control groups whose CD4 counts fall below a particular threshold. In panel (a), the threshold is 200, which is regarded as the level where AIDS can initiate. In panel (b), the threshold is 250. In panel (c), the threshold is 300.

Figure 5: Effects on Employment of the introduction of HAART treatment around the Time of HIV Diagnosis



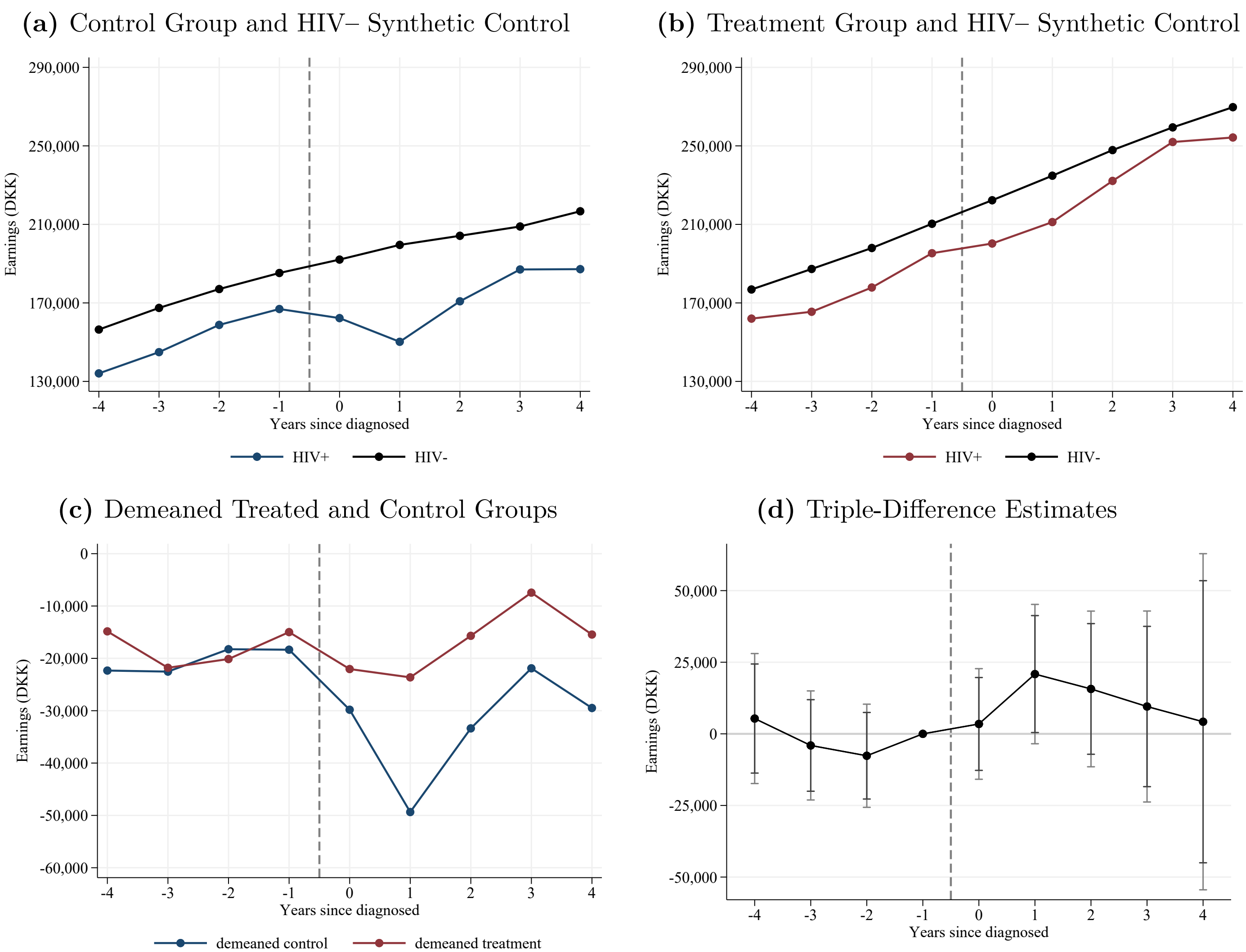
Notes: These graphs plot the effects on employment of the introduction of HAART treatment on individuals around the time when they are diagnosed with HIV. Graph (a) plots an event study for the control group (those diagnosed with HIV between 1990 and 1993, before the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (b) plots an event study for the treatment group (those diagnosed with HIV between 1995 and 1999, after the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (c) plots the control and treatment groups demeaned by their respective synthetic controls. Graph (d) plots the β_t estimates of the triple difference model estimated in Equation (2), which identify dynamic causal effects of the introduction of HAART treatment. We report 95% and 90% confidence intervals calculated from clustered standard errors.

Figure 6: Effects on Earnings of the introduction of HAART treatment around the Time of HIV Diagnosis



Notes: These graphs plot the effects on earnings of the introduction of HAART treatment on individuals around the time when they are diagnosed with HIV. Graph (a) plots an event study for the control group (those diagnosed with HIV between 1990 and 1993, before the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (b) plots an event study for the treatment group (those diagnosed with HIV between 1995 and 1999, after the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (c) plots the control and treatment groups demeaned by their respective synthetic controls. Graph (d) plots the β_t estimates of the triple difference model estimated in Equation (2), which identify dynamic causal effects of the introduction of HAART treatment. We report 95% and 90% confidence intervals calculated from clustered standard errors.

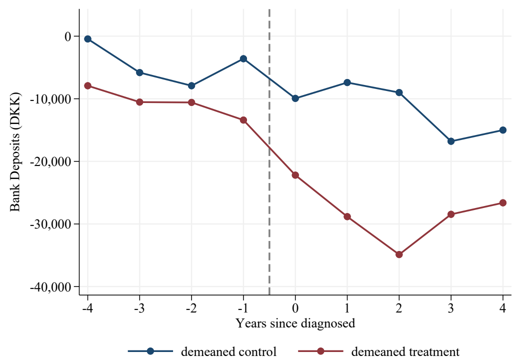
Figure 7: Effects on Earnings (Conditional on Employment)
of the introduction of HAART Treatment around the Time of HIV Diagnosis



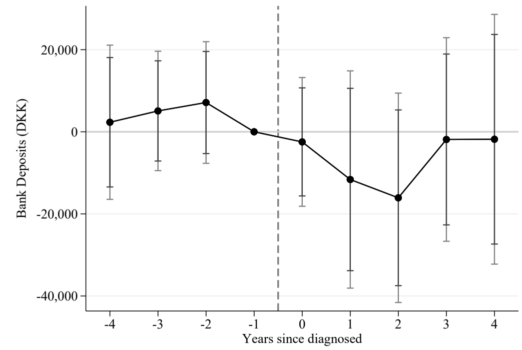
Notes: These graphs plot the effects on earnings conditional on participation of the introduction of HAART treatment on individuals around the time when they are diagnosed with HIV. Graph (a) plots an event study for the control group (those diagnosed with HIV between 1990 and 1993, before the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (b) plots an event study for the treatment group (those diagnosed with HIV between 1995 and 1999, after the introduction of HAART) against a synthetic control of HIV− individuals matched in year -4 from the same cohort, age, and gender. Graph (c) plots the control and treatment groups demeaned by their respective synthetic controls. Graph (d) plots the β_t estimates of the triple difference model estimated in Equation (2), which identify dynamic causal effects of the introduction of HAART treatment. We report 95% and 90% confidence intervals calculated from clustered standard errors.

Figure 8: Effects on wealth of the introduction of HAART treatment around the Time of HIV Diagnosis

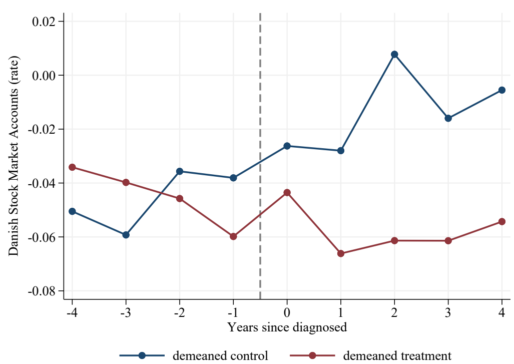
(a) Bank Deposits



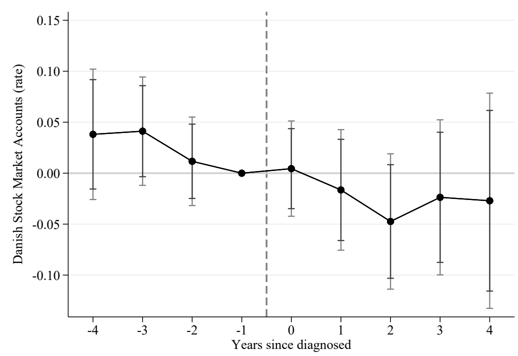
(b) Bank Deposits



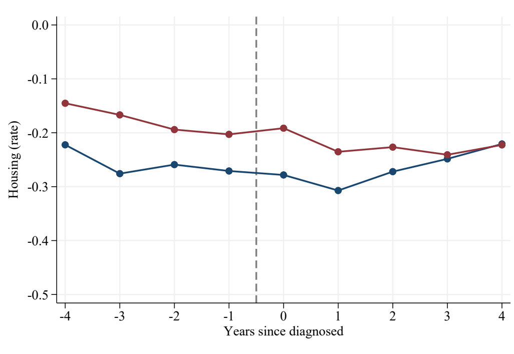
(c) Stock Ownership



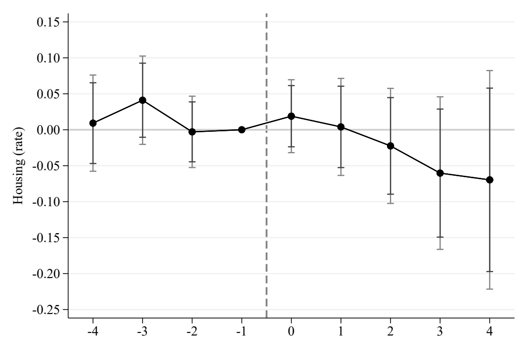
(d) Stock Ownership



(e) Housing Ownership

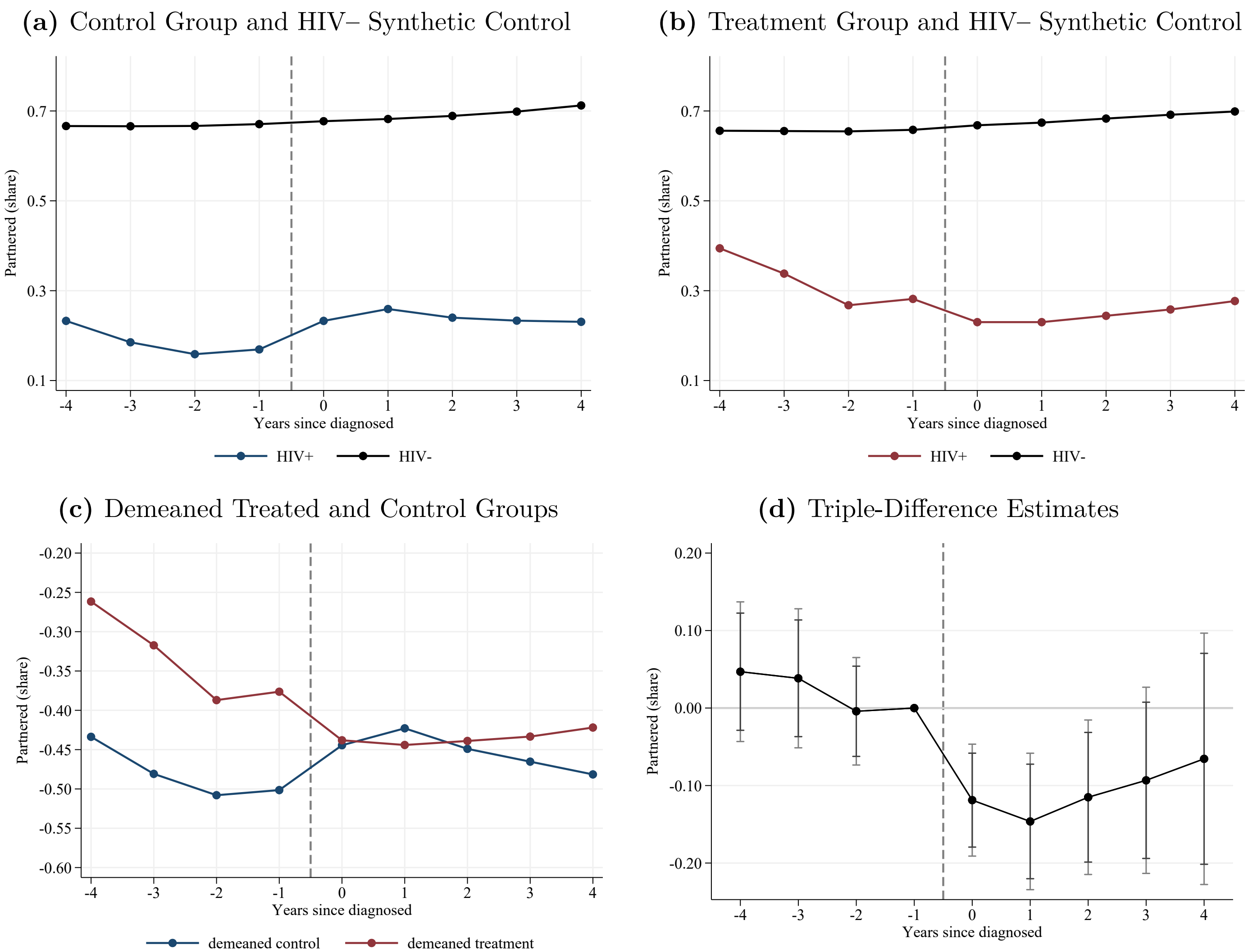


(f) Housing Ownership



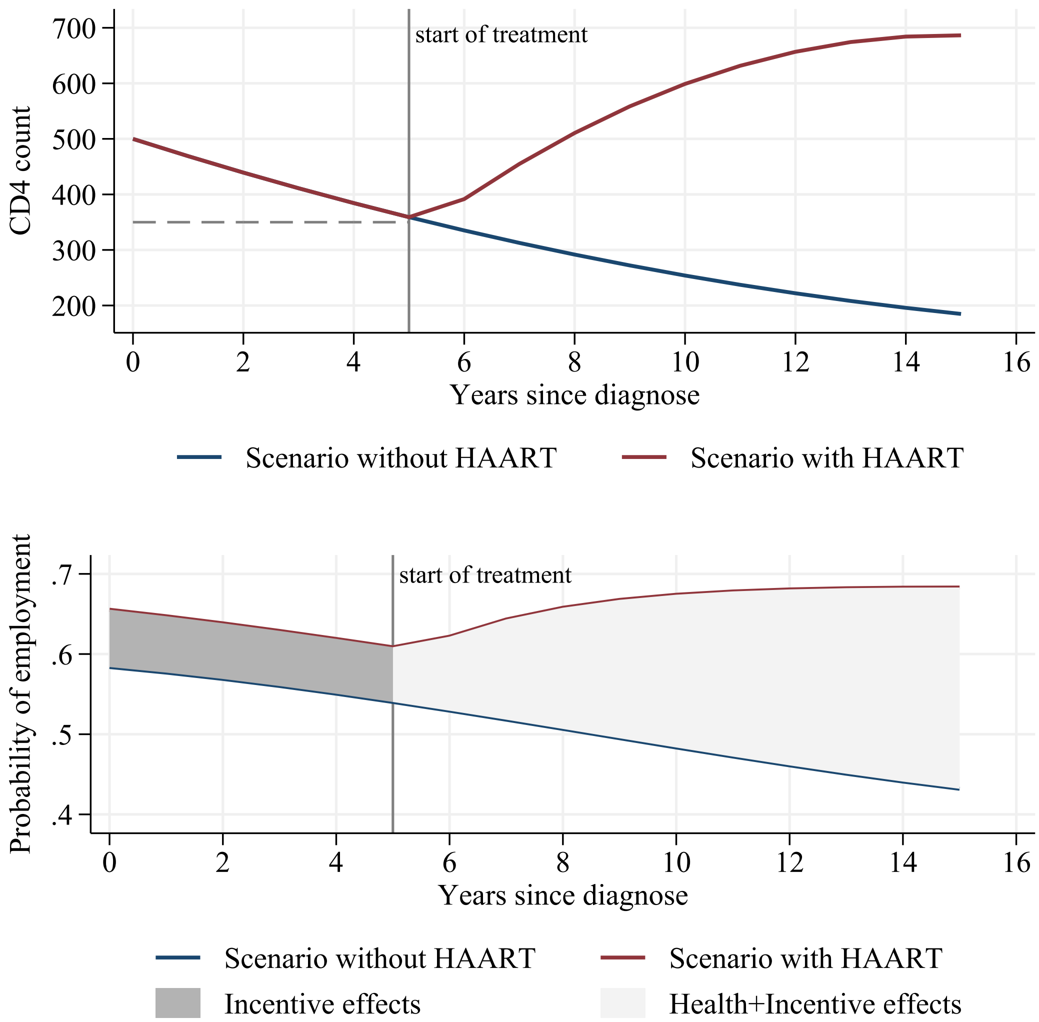
Notes: These graphs plot the effects on bank account savings, stocks, and housing of the introduction of HAART treatment on individuals around the time when they are diagnosed with HIV. Graphs (a, c, and e) on the left plot the control and treatment groups demeaned by their respective synthetic controls. Graphs (b, d, and f) on the right plot the estimates of the dynamic triple difference model estimated in Equation (2).

Figure 9: Effects on Partnership of the introduction of HAART treatment around the Time of HIV Diagnosis



Notes: These graphs plot the effects on marital status (being married or in cohabitation) of the introduction of HAART treatment on individuals around the time when they are diagnosed with HIV. Graph (a) plots an event study for the control group (those diagnosed with HIV between 1990 and 1993, before the introduction of HAART) against a synthetic control of HIV– individuals matched in year -4 from the same cohort, age, and gender. Graph (b) plots an event study for the treatment group (those diagnosed with HIV between 1995 and 1999, after the introduction of HAART) against a synthetic control of HIV– individuals matched in year -4 from the same cohort, age, and gender. Graph (c) plots the control and treatment groups demeaned by their respective synthetic controls. Graph (d) plots the β_t estimates of the triple difference model estimated in Equation (2), which identify dynamic causal effects of the introduction of HAART treatment. We report 95% and 90% confidence intervals calculated from clustered standard errors.

Figure 10: CD4 and Employment with and without HAART



Notes: Top panel: The figure is based on two fixed effect regressions of CD4 counts on time since diagnosis and time since treatment initiated, see appendix C, step 1. The lower panel is calculated based on estimates from a regression of employment on CD4 counts. We allow the effect to depend on whether HAART was available at diagnosis, see Appendix C.

Table 1: Descriptive Statistics

	Control (1)	Treated (2)	Difference (3)	*P-value* (4)	HIV– (5)
Demographics					
Age	33.4	34.5	-1.15	0.20	34.0
Male	0.84	0.80	0.04	0.26	0.82
Education (years)	11.58	11.77	-0.19	0.44	11.8
Dane	0.94	0.91	0.03	0.26	0.92
Economic outcomes					
Employed	0.69	0.68	0.01	0.79	0.79
Earnings (diff. HIV–)	-26,918	-27,701	783	0.95	164,270
Earnings (quartile)	2.19	2.29	-0.11	0.34	2.49
Home Owner	0.22	0.27	-0.05	0.29	0.47
Stocks Ownership (diff. HIV–)	-0.04	-0.06	0.02	0.46	0.15
Marital					
Married	0.11	0.18	-0.08	0.03	0.44
Cohabiting	0.06	0.10	-0.04	0.20	0.22
Health					
Hospital visit	0.15	0.14	0.01	0.73	0.18
Charlson Index	0.00	0.02	-0.02	0.12	0.010
Infections	<0.03	<0.03	-0.01	0.18	0.001
Psychologist/psychiatry	<0.03	<0.04	-0.01	0.41	0.010
DANHIV					
CD4 Count (cond. > 400)	615	620	-5.29	0.78	–
Heterosexual	0.39	0.44	-0.05	0.31	–
Observations	189	213			402,000

Notes: This table presents summary statistics of key variables for different samples, measured one year before diagnosis, except CD4 counts, which are measured in the year of diagnosis. Column (1) corresponds to the control group of the analysis sample: Individuals diagnosed with HIV between 1990 and 1993. Column (2) corresponds to the treatment group of the analysis sample: Individuals diagnosed with HIV between 1995 and 1999. Column (3) shows the difference in means between columns (1) and (2). Column (4) reports the *p-value* for a test of equal means. A joint test for all variables reported in the table cannot reject the hypothesis of equal means with a *p-value* of 0.44. Column (5) corresponds to a sample of individuals who are not diagnosed with HIV. This sample is constructed by matching 1,000 individuals of the same cohort, age, and gender to each of the individuals in the analysis sample. Variables marked as (diff. HIV–) are computed as the difference between each HIV+ individual with respect to their matched HIV– individuals to absorb calendar time effects. In these cases, column (5) reports the mean value of the HIV– individuals.

Table 2: Effects of Access to HAART on Labor and Wealth Outcomes. Triple-Difference Estimates

	Estimate (1)	Mean (2)
A: Labor Outcomes		
Employment	0.0551*	0.69
	(0.0328)	
Earnings	11,413	139,254
	(7,787)	
Earnings (cond. part.)	16,059*	181,755
	(8,774)	
B: Assets		
Bank Accounts	-12,495	28,648
	(8,012)	
Any Stocks	-0.0421	0.10
	(0.0261)	
Home Ownership	-0.0220	0.25
	(0.0310)	
Observations	3,081,078	
N. Individuals	402,402	

Notes: Column (1) in this table reports the coefficient of interest β_1 estimated in Equation (3) that captures the causal effect of the introduction of HAART medical innovation that extended life expectancy on different outcomes, up to 3 years following diagnosis. Column (2) reports the average value of a given outcome measured the year before HIV diagnosis for the sample of analysis. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.1$

Table 3: Marital Effects of the introduction of HAART Treatment after HIV Diagnosis. Triple-Difference Estimates.

	Full Sample (1)	Hetero. (2)	Homo. (3)	Mean (4)
Partnered	-0.142***	-0.154**	-0.144***	0.23
	(0.0369)	(0.0621)	(0.0439)	
Married	-0.0796**	-0.0705	-0.0922**	0.15
	(0.0320)	(0.0527)	(0.0395)	
Cohabiting	-0.0625***	-0.0835*	-0.0515**	0.08
	(0.0232)	(0.0432)	(0.0237)	
Divorce	0.00236	-0.0134	0.00926	0.02
	(0.0109)	(0.017)	(0.0135)	
Observations	3,081,078	1,404,403	1,676,675	3,081,078
N. Individuals	402,402	184,184	218,218	402,402

Notes: Columns (1) to (3) in this table report the coefficient of interest, β_1, estimated in Equation (3), that captures the causal effect of the introduction of HAART treatment that extended life expectancy on different outcomes, for the full sample as well as distinguishing by sexual orientation, which is inferred from the stated source of HIV infection. Note that the cohabitation outcome is constructed based on opposing-sex partnerships, also in column (3). Column (4) reports the average value of a given outcome measured the year before HIV diagnosis for the full sample. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.1$

Table 4: Health Effects of the introduction of HAART Treatment after HIV Diagnosis. Triple-Difference Estimates

	Estimate (1)	Mean (2)
A: Physical Health		
Hospital visit	0.0172	0.14
	(0.0294)	
Charlson Index	0.00455	0.01
	(0.0103)	
Infections	0.00291	<0.03
	(0.00441)	
B: Mental Health		
Psychologist/Psychiatry	-0.0131	<0.03
	(0.0137)	
Observations	3,081,078	
N. Individuals	402,402	

Notes: Column (1) in this table reports the coefficient of interest β_1 estimated in Equation (3) that captures the causal effect of the introduction of HAART medical innovation that extended life expectancy on different outcomes, up to 3 years following diagnosis. Column (2) reports the average value of a given outcome measured the year before HIV diagnosis for the sample of analysis. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.1$

Table 5: Education Effects of the introduction of HAART Treatment after HIV Diagnosis. Triple-Difference Estimates

	Estimate (1)	Mean (2)
Years of Education	-0.044	11.6
	(0.081)	
Observations	3,081,078	
N. Individuals	402,402	

Notes: Column (1) in this table reports the coefficient of interest β_1 estimated in Equation (3) that captures the causal effect of the introduction of HAART medical innovation that extended life expectancy on different outcomes, up to 3 years following diagnosis. Column (2) reports the average value of a given outcome measured the year before HIV diagnosis for the sample of analysis. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$, $^{*}\, p < 0.1$

Table 6: Robustness to alternative sample of analysis or specification choice

	Employment			Assets			Marital				Health			
	Empl.	Earn.	Earn.P.	Bnk.Ac.	Stocks	Hous.	Part.	Marr.	Coh.	Div.	Hosp.	Char.	Inf.	Psych.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
A. Baseline	0.0551*	11413.0	16059.1*	-12494.7	-0.0421	-0.0220	-0.142***	-0.0796**	-0.0625***	0.00236	0.0172	0.00455	0.00291	-0.0131
	(0.0328)	(7787.2)	(8773.7)	(8011.5)	(0.0261)	(0.0310)	(0.0369)	(0.0320)	(0.0232)	(0.0109)	(0.0294)	(0.0103)	(0.00441)	(0.0137)
B. Controlling for CD4	0.0551*	11413.0	16127.5*	-12494.7	-0.0421	-0.0220	-0.142***	-0.0796**	-0.0625***	0.00236	0.0172	0.00455	0.00291	-0.0131
	(0.0328)	(7787.6)	(8779.7)	(8010.6)	(0.0261)	(0.0310)	(0.0369)	(0.0320)	(0.0232)	(0.0109)	(0.0294)	(0.0103)	(0.00441)	(0.0137)
C. Controlling for year of diagnosis	0.0551*	11413.0	16816.9*	-12494.7	-0.0421	-0.0220	-0.142***	-0.0796**	-0.0625***	0.00236	0.0172	0.00455	0.00291	-0.0131
	(0.0328)	(7794.8)	(8743.4)	(8012.2)	(0.0261)	(0.0310)	(0.0369)	(0.0320)	(0.0232)	(0.0109)	(0.0294)	(0.0103)	(0.00441)	(0.0137)
D. With 1994 diagnoses	0.0546*	11193.6	15011.2*	-11961.1	-0.0357	0.00108	-0.116***	-0.0603*	-0.0562**	0.000779	0.0134	0.00619	0.00277	-0.0108
	(0.0319)	(7693.2)	(8576.2)	(7779.9)	(0.0254)	(0.0305)	(0.0367)	(0.0319)	(0.0233)	(0.0112)	(0.0289)	(0.0103)	(0.00431)	(0.0135)
E. Excluding if received HAART	0.0896**	17161.9*	17306.7	-13807.7	-0.0595**	-0.0633*	-0.185***	-0.0931**	-0.0917***	-0.00125	0.00558	0.00494	0.00202	-0.0146
	(0.0423)	(9430.8)	(10632.3)	(9051.8)	(0.0303)	(0.0373)	(0.0490)	(0.0431)	(0.0290)	(0.0144)	(0.0370)	(0.0135)	(0.00616)	(0.0173)

Notes: This table reports the coefficient of interest for all main outcome variables considered in the analysis under alternative analysis samples or specification choices. Row A reports our baseline estimation (as reported on Tables 2, 3 and 4). Row B controls for CD4. Row C controls for the year of diagnosis. Row E drops individuals if they ever receive HAART during the analysis period. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table 7: Heterogeneity analysis by age, gender, and education

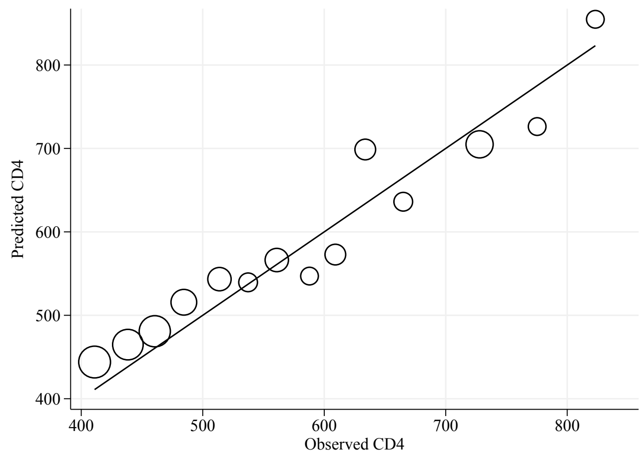
	Employment			Assets			Marital			
	Empl. (1)	Earn. (2)	Earn.P. (3)	Bnk.Ac. (4)	Stocks (5)	Hous. (6)	Part. (7)	Marr. (8)	Coh. (9)	Div. (10)
A: Age										
Baseline group (young)	0.0399	8057.2	13280.1	-3429.3	-0.0602*	-0.0384	-0.154***	-0.0892*	-0.0647*	0.00457
	(0.0509)	(10859.9)	(11354.8)	(3724.1)	(0.0313)	(0.0434)	(0.0578)	(0.0464)	(0.0386)	(0.0151)
Interaction (old)	0.0255	3186.0	318.5	-14407.9	0.0333	0.0323	0.0233	0.0243	-0.000955	-0.00448
	(0.0736)	(18855.0)	(19942.0)	(16946.9)	(0.0535)	(0.0676)	(0.0753)	(0.0660)	(0.0454)	(0.0222)
Mean of baseline	0.691	116133.9	150006.3	12222.5	0.0922	0.166	0.230	0.129	0.101	0.0276
Mean of interaction group	0.681	166374.7	219852.3	47914.8	0.114	0.341	0.227	0.168	0.0595	0.0162
B: Gender										
Baseline group (male)	0.0572	15953.3*	18498.5*	-8951.9	-0.0324	-0.0190	-0.107***	-0.0693**	-0.0379*	0.00563
	(0.0358)	(8852.3)	(9772.0)	(7603.9)	(0.0305)	(0.0352)	(0.0378)	(0.0334)	(0.0220)	(0.0114)
Interaction (female)	-0.0122	-24155.6	-16073.6	-26432.2	-0.0468	-0.0176	-0.195	-0.0632	-0.132	-0.0181
	(0.0930)	(17399.0)	(20373.0)	(32940.5)	(0.0494)	(0.0710)	(0.119)	(0.101)	(0.0848)	(0.0343)
Mean of baseline	0.720	149817.7	191790.0	33068.8	0.116	0.264	0.201	0.134	0.0669	0.0122
Mean of interaction group	0.534	91648.6	131183.3	8724.5	0.0411	0.164	0.356	0.205	0.151	0.0685
C: Education										
Baseline group (low educ)	0.0445	14908.3	29959.2*	-5817.9	-0.00266	-0.0189	-0.228***	-0.106**	-0.122***	0.00857
	(0.0522)	(11494.9)	(15477.9)	(11721.6)	(0.0311)	(0.0400)	(0.0597)	(0.0535)	(0.0421)	(0.0190)
Interaction (high educ)	0.0150	-5611.5	-21291.0	-10924.1	-0.0662	-0.00525	0.147*	0.0455	0.102**	-0.00998
	(0.0669)	(15608.8)	(18809.0)	(15794.2)	(0.0515)	(0.0590)	(0.0758)	(0.0684)	(0.0492)	(0.0231)
Mean of baseline	0.586	107117.8	166509.9	24255.4	0.0764	0.153	0.274	0.178	0.0955	0.0255
Mean of interaction group	0.751	159848.5	189192.7	31462.9	0.118	0.306	0.200	0.127	0.0735	0.0204

Notes: This table reports baseline and interaction coefficients across different heterogeneity dimensions based on a binary categorization. The effect on the baseline group corresponds to the reported coefficient for the baseline group. The effect on the complementary group corresponds to the baseline coefficient plus the reported interaction coefficient. Older group (those above 34 years old) represents 49.5%. Females represent 18.3%. The highly educated group represents 60.4%. $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.1$.

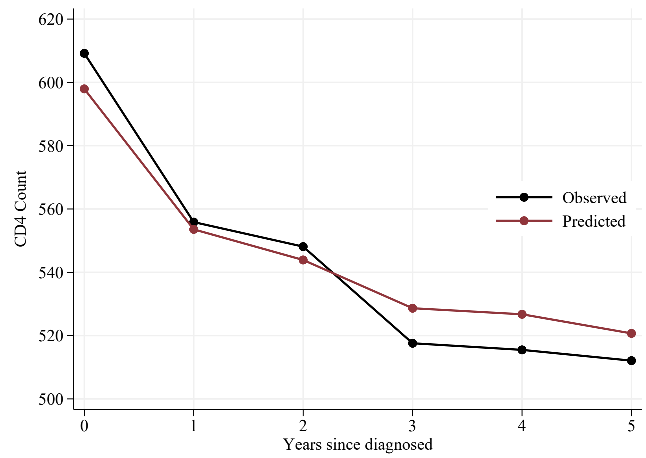
Appendix A Supplementary Figures and Tables

Figure A.1: Accuracy of the CD4 imputation model

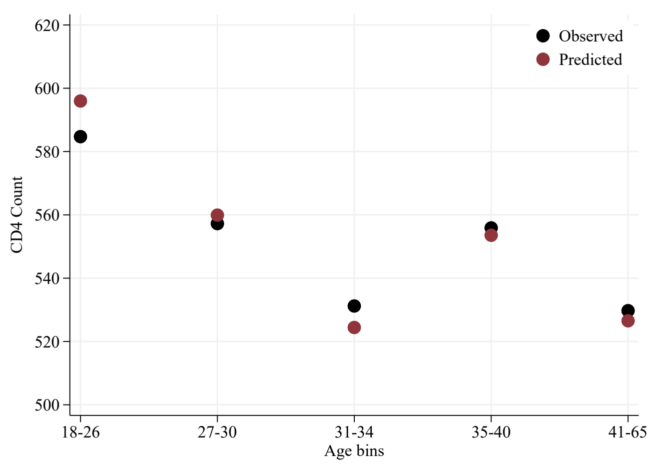
(a) By CD4 count level



(b) By time since diagnosis

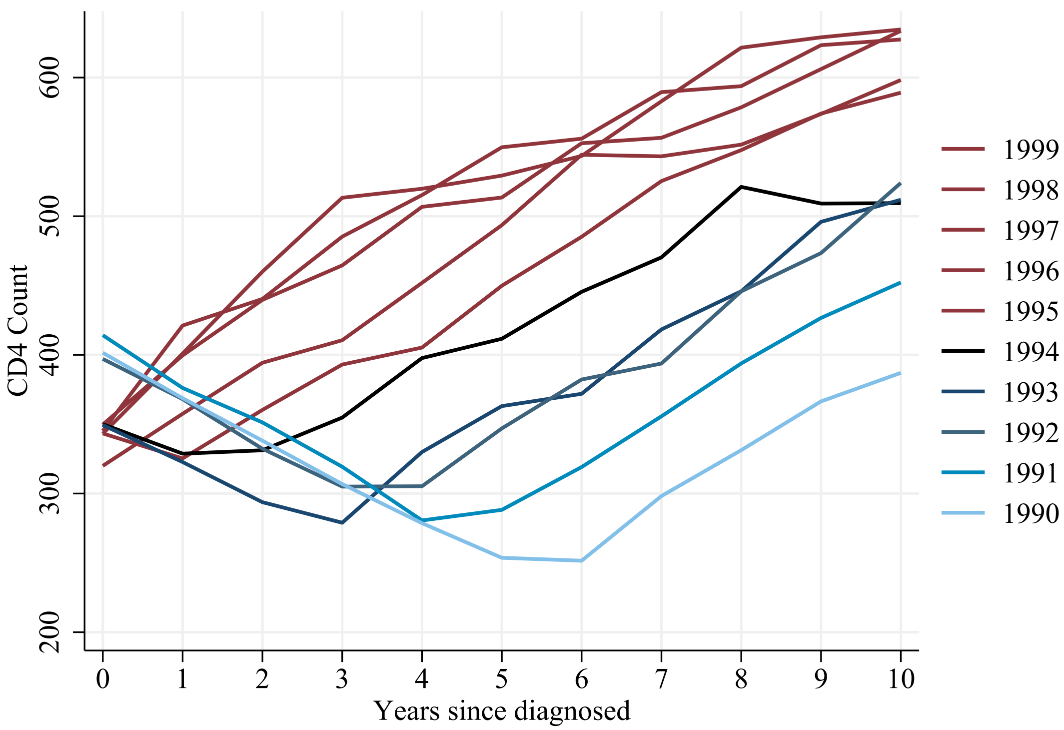


(c) By age



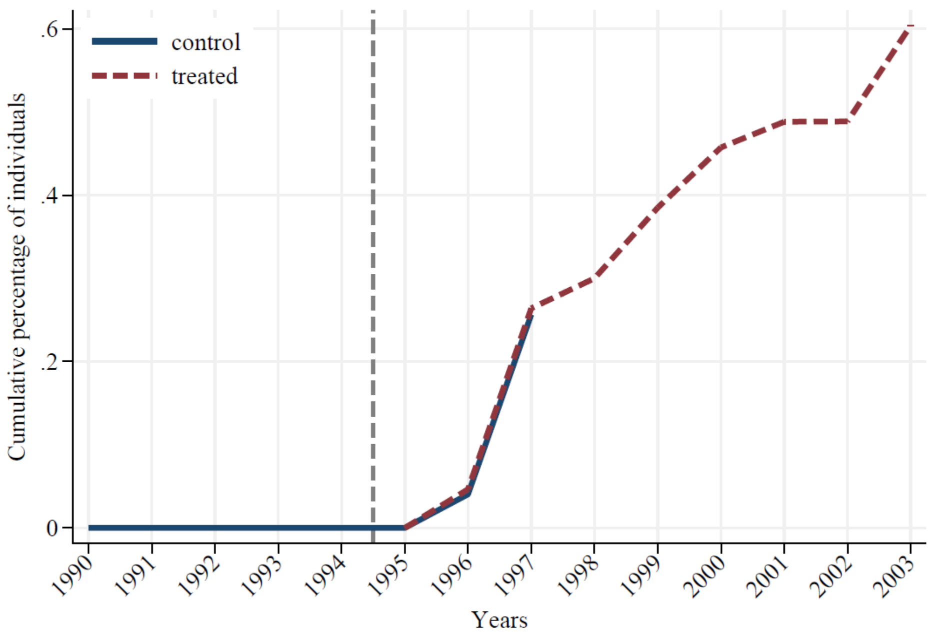
Notes: These graphs plot the observed CD4 count and the predicted CD4 count based on the model described in Equation (1) for the sample of individuals for whom we can observe CD4 counts. Specifically, the graphs show the sample of individuals diagnosed between 1995 and 1999 with a CD4 count above 400 at the time of diagnosis. Graph (a) shows the average CD4 count predicted by level of observed CD4 count at the time of diagnosis. Graph (b) shows the average CD4 count observed and predicted each year since diagnosis. Graph (c) shows the average CD4 count observed and predicted by equally sized age groups, as age is one of the few predictors identified in the medical literature as predictors of CD4 progression (Jia et al., 2020; Glaubius et al., 2021).

Figure A.2: Evolution of CD4 count after diagnosis, unrestricted sample



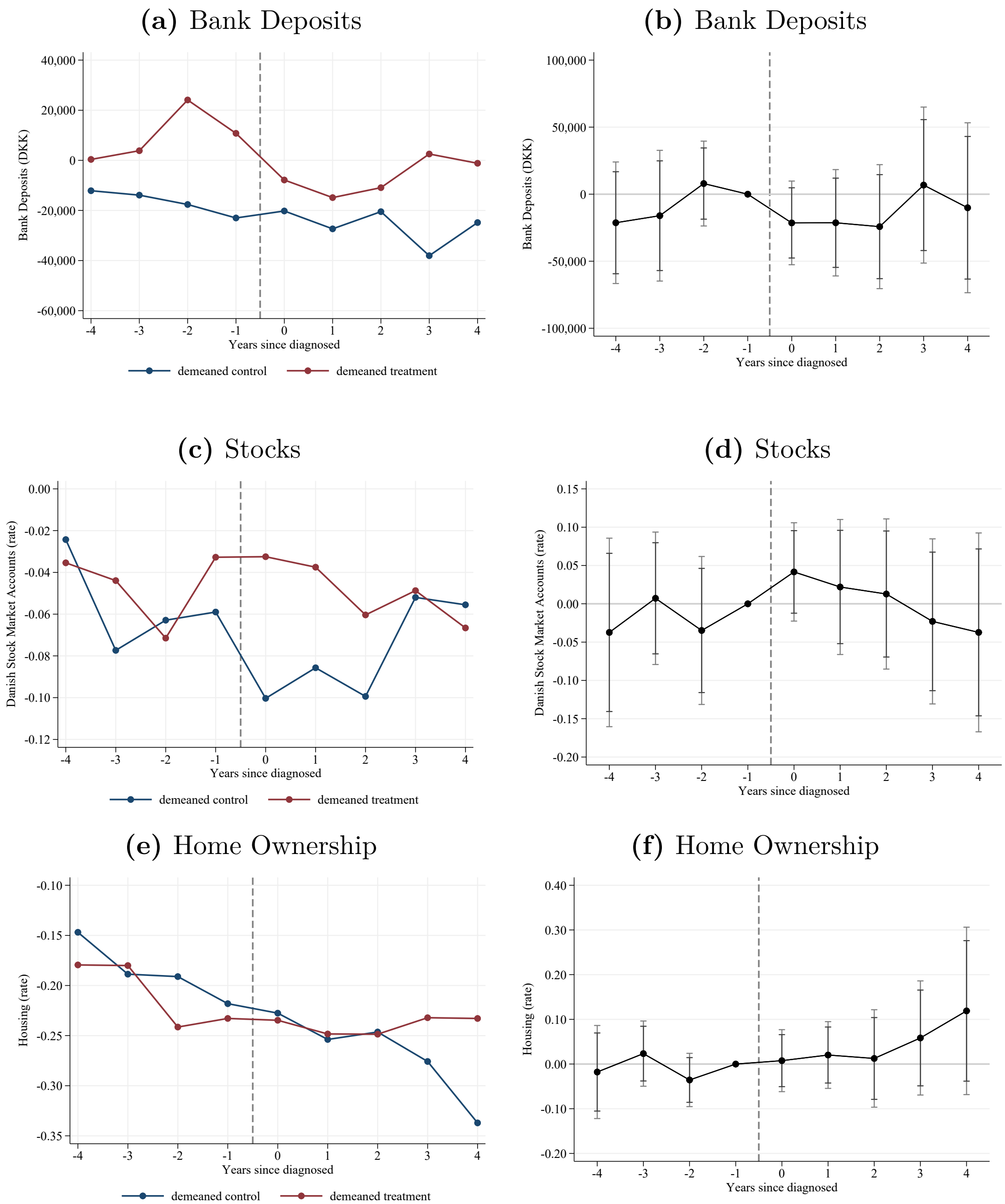
Notes: This figure shows the evolution of CD4 counts for different groups of individuals according to the year when they were diagnosed. The analysis is based on the unrestricted sample that includes all individuals infected with HIV, rather than on our analysis sample that is restricted to individuals diagnosed when their CD4 count levels are above 400. The graph illustrates how the introduction of HAART medication in 1995 helped reverse the deterioration of CD4 count levels.

Figure A.3: Take-up of HAART medication for the analysis sample



Notes: This figure shows, for our analysis sample, the cumulative share of individuals who have begun HAART treatment over time. Individuals in the control group are diagnosed between 1990 and 1993, while individuals in the treatment group are diagnosed between 1995 and 1999. For this figure, observations of the control group beyond 1995 are kept.

Figure A.4: Wealth outcomes for sample of individuals diagnosed with a low CD4 count (150-250)



(a) Bank Deposits **(b)** Bank Deposits

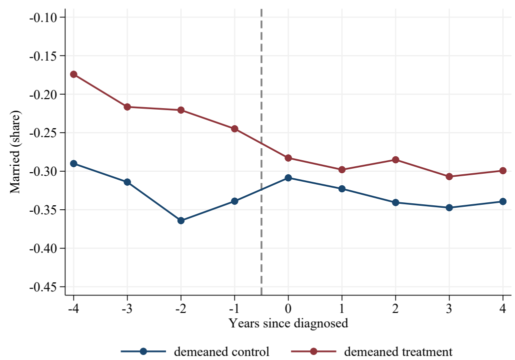
(c) Stocks **(d)** Stocks

(e) Home Ownership **(f)** Home Ownership

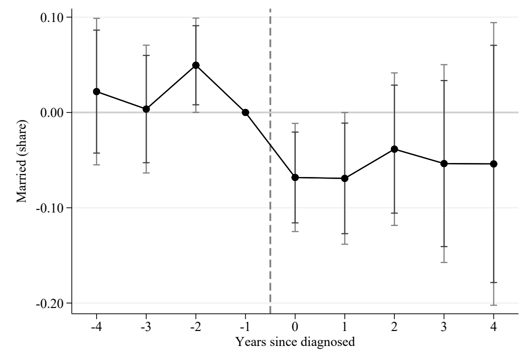
Notes: This figure shows estimates of wealth effects for a group of individuals with a lower CD4 count (150-250) than the main sample. Graphs (a, c, and e) on the left plot the control and treatment groups demeaned by their respective synthetic controls. Graphs (b, d, and f) on the right plot the estimates of the dynamic triple difference model estimated in Equation (2)

Figure A.5: Effects on Marital outcomes of the introduction of HAART Treatment around the Time of HIV Diagnosis on Other Marital Outcomes

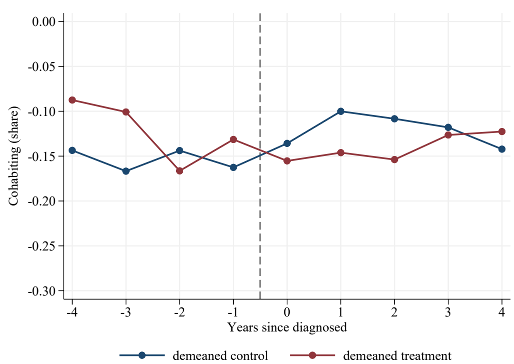
(a) Married



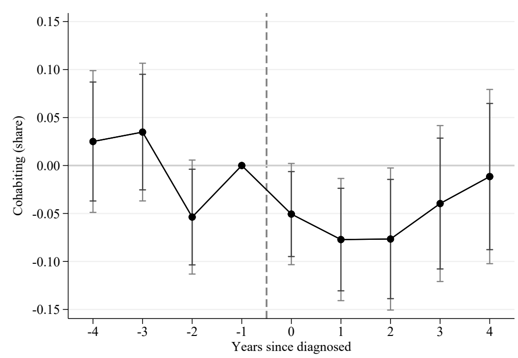
(b) Married



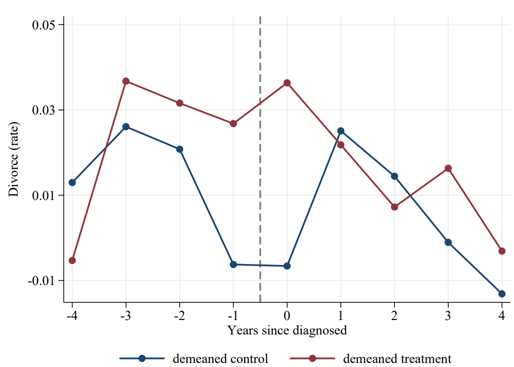
(c) Cohabitation



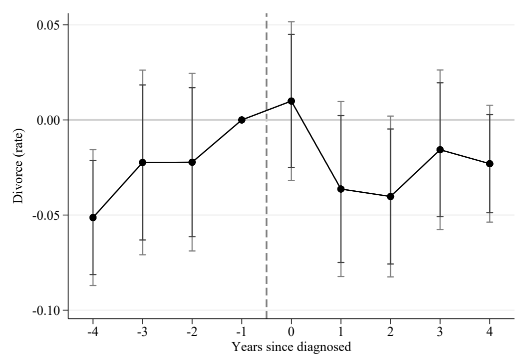
(d) Cohabitation



(e) Divorce



(f) Divorce



Notes: These graphs plot the effects on marriage, cohabitation, and divorces of HAART treatment on individuals around the time when they are diagnosed with HIV. Graphs (a, c, and e) on the left plot the control and treatment groups demeaned by their respective synthetic controls. Graphs (b, d, and f) on the right plot the estimates of the dynamic triple difference model estimated in Equation (2)

Appendix B The dissemination of news on the HAART treatment

The HAART medication was introduced in 1996 in Denmark, but the first positive indications of the new treatment were presented already in 1995. In the mid-1990s, substantial attention was given to the medical innovations in HIV treatment, and the media were frequently reporting from scientific conferences and events. In Table B.1, we show some important dates of the medical breakthroughs and examples of how the information was disseminated to a wider audience.

Anecdotal evidence suggests that the optimism among Danish researchers started after the "Fifth European Conference on Clinical Aspects and Treatment of HIV Infection" held on September 26-29, 1995, in Copenhagen. The positive news was disseminated to the Danish population already the day after the conference. An article published on September 30, 1995, in the Danish newspaper Politiken had the headline "Great confidence in new HIV medicine" by Kaare Skovmand. A quote translated from the article illustrates the growing optimism and the beginning of a new era with an effective treatment of HIV:

> *"The AIDS conference in Copenhagen gave international researchers a rare opportunity to bring out the smile. For the first time in many years, definite positive results could be presented, as several studies independently showed that many HIV-positive people can look forward to a longer life by being treated with a combination of the old drug AZT and the two newer drugs ddl and ddC."*

"Stor tiltro til ny HIV-medicin", Politiken September 30, 1995.

Table B.1: Development of HAART Treatment

Date	**Event**	**Example of News Coverage**
June 1995	FDA approves saquinavir, the first protease inhibitor. It becomes a key component of HAART treatment.	–
September 1995	Fifth European Conference on HIV Treatment, Copenhagen. Positive results for protease inhibitors presented.	Politiken, Sept 30: *Great confidence in new HIV medicine* by Kaare Skovmand.
December 1995	FDA approves use of saquinavir in combination with other drugs.	New York Times, Dec 8: *FDA backs a new drug to fight AIDS.*
July 1996	11th International AIDS Conference, Vancouver. HAART shown to be effective.	New York Times, July 15: *From the AIDS Conference: Talk about Life, Not Death.*

Notes: Own elaboration

Appendix C Calculations of incentive effects

In this section, we present a "back-of-the-envelope" calculation of the benefits from introducing a new medical treatment. The calculations illustrate that the benefits are underestimated if the incentive effects prior to the actual treatment are not accounted for. Most cost-benefit analyses begin to calculate the benefits when the treatment is started (see e.g. guidelines from National Institute for Health and Care Excellence, 2025)[24] or they assume that the treatment only affects evolution of CD4 counts, not cost-benefit associated with a certain level of CD4 counts (see Davies et al., 1999).[25]

In this analysis, we use data on all HIV patients diagnosed from 1986-2005 who were still alive in 1995, excluding drug addicts. We exclude extreme observations with CD4 counts above 1600.[26] Note that in this exercise, we use the actual CD4 counts and not the imputed ones. As in the main analysis, we use a matched control sample of individuals who are not infected with HIV. For each HIV patient, we include 1000 individuals with the same gender, year of birth, and observed in the same calendar years as the HIV-infected individuals are observed. The data contains 19,275 observations from 2,555 individuals who are infected with HIV and 19,275,000 observations from 2,555,000 individuals not infected with HIV (in the period before 1999). Table C.1 shows the descriptive statistics of the two samples. We observe that HIV-infected individuals are, on average, less employed, earn less, and are less likely to have a partner compared to non-infected individuals. Infected individuals also have a higher use of health services. Note that these results differ somewhat from our main sample of analysis described in Table 1, as expected given that we do not condition on having high CD4 counts at the time of diagnosis. Infected individuals in Table 1 have higher CD4 count levels by construction and are more similar to

[24]The guidelines (National Institute for Health and Care Excellence, 2025) do not mention any potential benefits before treatment is initiated.

[25]Davies et al. (1999) calculate the direct hospital costs associated with certain CD4 intervals, but they do not allow these costs to depend on whether or not treatment is available. The cost saving of the treatment in this exercise is obtained because the treatment affects the transition probabilities between the different CD4 intervals.

[26]We drop 22 observations out of more than 19.000 observations.

their matched non-infected sample in terms of health indicators, employment, and partnership.

The calculation is made in three steps.

Step 1: We estimate the development of CD4 counts before the HIV patient receives HAART treatment and the development after receiving HAART treatment. In this step, we only use the subsample of HIV-infected individuals (columns (1) and (2) in Table C.1). We do this by estimating two separate fixed effect regressions of CD4 counts: one for individuals who are diagnosed but have not yet received HAART treatment, and one for individuals after HAART treatment has started. The estimation results are shown in the following equations:

$$\begin{aligned} CD4_{it} &= \underset{(5.07)}{518.70} - \underset{(1.55)}{30.02} \cdot ysd_{it} + \underset{(0.10)}{0.65} \cdot ysd_{it}^2 + \widehat{\alpha_i} + \widehat{\epsilon_{it}} \\ CD4_{it} &= \underset{(2.31)}{263.12} + \underset{(1.36)}{72.06} \cdot ysh_{it} - \underset{(0.16)}{3.83} \cdot ysh_{it}^2 + \widehat{\alpha_i} + \widehat{\epsilon_{it}}, \end{aligned} \tag{4}$$

where *ysd* is year since diagnosis and *ysh* is years since HAART treatment was initiated.

Step 2: We estimate a flexible function for how the CD4 counts affect employment. We use the sample that includes also HIV− individuals in this exercise and assume that individuals who have not been infected with HIV have a CD4 count of 1000.[27] In the specification, we allow the impact of CD4 to depend on whether individuals were diagnosed before or after the introduction of HAART. We use the following model:

$$\begin{aligned} Empl_{it} =& \gamma_0 + \delta_0 \cdot Treat_i + X_{it}\beta + \lambda_1(1 - Inf_i) \\ &+ \sum_{s=1}^{3} \gamma_s (CD4_{it})^s + \sum_{s=1}^{3} \delta_s (CD4_{it})^s \cdot Treat_i + u_{it}, \end{aligned} \tag{5}$$

where X contains age, months of education, a dummy for male, a dummy for being Dane, and calendar year dummies. Inf_i is a dummy for individual i if infected

[27]The estimation is not sensitive to the value of CD4 of non-infected individuals, as we include a control for non-infected individuals.

with HIV, and $Treat$ is a dummy for being diagnosed after treatment is available.[28] In order to make the estimation comparable to the main estimation, the treatment dummy $Treat$ is defined as described in Section 4, which implies that we leave out the individuals who are diagnosed in 1994 as well as their matched controls. The estimation results are shown in Table C.2, column (1).

Based on the estimates in Table C.2, we construct the predicted employment rate for a Danish man who is 34 years old in 1995 and has 143 months of education. The results, shown in Figure C.1, confirm that even at the same level of CD4 counts, the cohorts diagnosed after the HAART treatment was available are more likely to work. The estimated effect of having access to treatment for healthy individuals (CD4>500) is estimated to be 7.2 percentage points (see Table C.2 column (1)). The estimate is slightly higher than the estimated effect of 5.5 percent point obtained from the triple difference estimator (see Table 2).[29] Furthermore, the figure shows that as health (and CD4) declines, the probability of working is also declining.

To check the robustness of these results, we perform additional robustness analyses. First, we estimate the regression model on a restricted sample where we exclude observations from the control cohorts after 1995 (when treatment was also available for the control cohorts). The results are shown in column (2) in Table C.2. Second, we estimate the model using the imputed CD4 count instead of the actual CD4 count. In column (3), the results with imputed CD4 are shown for the full sample, and in column (4) for the restricted sample. The results show that the estimated effect of treatment is, in all cases, positive and significant. The estimates are in the range of $5-9$ percentage points, indicating that an HIV-infected individual is between $5-9$ percentage points more likely to work if treatment is available compared to when

[28]For comparison, we also estimate a specification with indicator functions:

$$\begin{aligned} Empl_{it} =& \gamma_0 + \delta_0 \cdot Treat_i + X_{it}\beta + \lambda_1(1 - Inf_i) \\ &+ \sum_s \gamma_s \mathbb{1}(CD4_{it} \in I_s) + \sum_s \delta_s \mathbb{1}(CD4_{it} \in I_s) \cdot Treat_i + u_{it}, \end{aligned} \tag{6}$$

where $\mathbb{1}$ is an indicator function, and I_s is an interval of size 50.

[29]A potential explanation for this slightly higher estimate is that it is based on a longer period after diagnosis (6.5 years on average), while the estimate in Table 2 is based on four years. The dynamic results in Figure 5 show higher effects on the later years.

treatment was not available.

Step 3: We calculate the predicted probability of employment and compare the predictions for an individual in the two scenarios, one where HAART treatment exists and one where HAART does not exist. We assume that HAART treatment is initiated when the CD4 counts fall below 350, as the guidelines stipulated in the 1990s. To illustrate the benefits of the new medical innovation, we calculate the probability of employment for a Danish man aged 34 who has 143 months of education and has CD4 counts equal to 500 at the time of the diagnose.[30]

We use the estimated equations in Equation (4) to predict the development of CD4 in the two scenarios, as shown in top panel of Figure 10.[31] The HAART treatment in the scenario with HAART will be initiated 5 years after diagnosis, since the CD4 will be below 350. Based on the (predicted) CD4 counts, we show in Figure C.1 the predicted employment. In this exercise, we use the specification for the polynomial, but the results are almost unchanged if we use the dummy approach instead.

The lower panel in Figure 10 shows that even before individuals receive the treatment (whuile CD4 counts are similar in both scenarios), employment in the scenario where HAART has been discovered is higher. The difference is due to the incentive effect of individuals who already anticipate that they will have access to the treatment in the future (dark gray area). We quantify the incentive effect as the dark gray area, which is equivalent to 0.49 years of employment during the first 5 years after diagnosis.

The difference between scenarios after the treatment has started is due to both a health and an incentive effect (both represented by the light gray area, which is equivalent to 1.83 years of employment). The total employment benefit of the HAART innovation should, therefore, be the sum of the dark gray and the light gray areas.

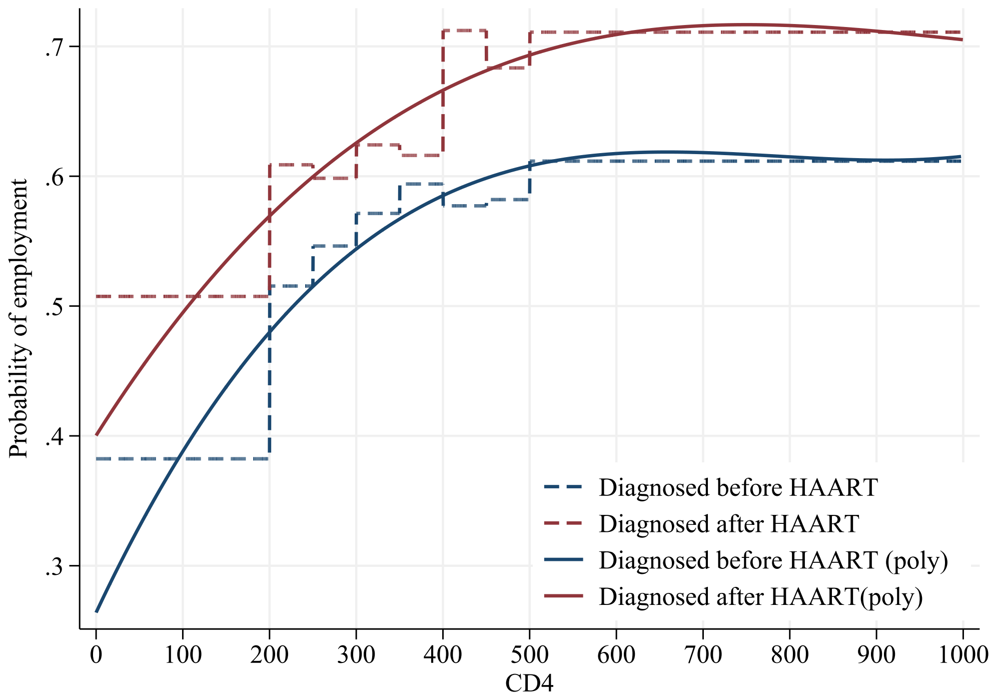
[30]Studies that evaluate the cost effectiveness of HAART treatment often use a CD4 count of 500 (or above) as the initial value of CD4 count for HIV patients, see Davies et al. (1999) and Freedberg et al. (2001).

[31]Our estimates of Equation (4) indicate a decline in CD4 of about 30 per year when HAART treatment is not initiated. This implies that a person who has a CD4 count of 500 at diagnosis would reach a CD4 count below 200 after about 10 years, at which point it is likely to develop AIDS and live for about two more years.

In this example, the total difference in employment between those with access to HAART and those without is 2.33 years of employment during the first 15 years after diagnosis. This calculation indicates that the total employment effect of HAART treatment during the first 15 years is underestimated by about $0.49/(1.83+0.49) =$ 21 percent if the incentive effect is not accounted for. This back-of-the-envelope calculation demonstrates that the full benefits of introducing medical technologies, such as HAART, may be underestimated unless both types of effects are accounted for.

Our results for labor supply illustrate the importance of accounting for the incentive effect arising before the onset of treatment when assessing the benefits. The gains from the incentive effects are, in general, important if there is uncertainty about the availability of appropriate treatment. This uncertainty can arise because of a lack of technological innovation or if there is uncertainty about eligibility for treatment, e.g., insurance coverage.

Figure C.1: Employment and CD4 counts, unrestricted sample



Notes: This figure shows estimates of the impact of CD4 counts on employment. The estimated effects are calculated for a man aged 34 with 143 months of education. The year is set to 1995. The blue curve refers to the estimates for a patient without access to the HAART treatment, while the red curve is for a patient with access to HAART treatment. The dashed lines refer to the estimates obtained from an OLS regression of Equation (5) while the solid lines refer to Equation (6).

Table C.1: Descriptive Statistics of the sample for the back-of-the-envelope calculation

	Infected		Non-infected	
	mean	sd	mean	sd
	(1)	(2)	(3)	(4)
Demographics				
Age	40.971	9.736	40.971	9.736
Male	0.779	0.415	0.779	0.415
Education (years)	12.16	2.67	12.42	2.69
Dane	0.769	0.421	0.938	0.240
Economic outcomes				
Employed	0.493	0.500	0.775	0.418
Earnings	120,681	159,156	217,053	188,938
Marital				
Cohabiting	0.317	0.465	0.722	0.448
Married	0.260	0.439	0.545	0.498
Health				
Hospital visit	0.342	0.475	0.285	0.451
Charlson Index	0.053	0.344	0.021	0.228
Infections	0.004	0.060	0.002	0.042
Psychologist/psychiatry	0.029	0.168	0.015	0.120
DANHIV				
CD4 Count	429	264	.	.
Heterosexual	0.386	0.487	.	.
Observations	19,275		19,275,000	

Notes: This table presents summary statistics of key variables for the sample used for the back-of-the-envelope calculation. Columns (1) and (2) correspond to the sample of individuals infected by HIV in the period 1985-1999. Columns (3) and (4) correspond to the matched sample of non-infected individuals. This sample is constructed by matching 1,000 individuals of the same cohort, age, and gender to each of the individuals in the sample of columns (1) and (2).

Table C.2: Estimation results: employment effects

	Actual CD4		Imputed CD4	
	Full sample	Restricted sample	Full sample	Restricted sample
	(1)	(2)	(3)	(4)
Age	-0.008***	-0.008***	-0.007***	-0.007***
	(0.000)	(0.000)	(0.000)	(0.000)
Male	0.030***	0.030***	0.027***	0.027***
	(0.000)	(0.000)	(0.000)	(0.000)
Education in months	0.002***	0.002***	0.002***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)
Dane	0.234***	0.234***	0.208***	0.208***
	(0.000)	(0.000)	(0.000)	(0.000)
Treatment	0.072***	0.088*	0.080***	0.052***
	(0.013)	(0.035)	(0.009)	(0.012)
Non-infected	0.173***	0.189***	0.190***	0.162***
	(0.010)	(0.034)	(0.006)	(0.009)
CD4(0-200)	-0.260***	-0.304***	-0.167***	-0.154***
	(0.016)	(0.041)	(0.008)	(0.012)
CD4(201-250)	-0.134***	-0.197***	-0.047***	-0.017
	(0.024)	(0.059)	(0.012)	(0.017)
CD4(251-300)	-0.059*	-0.167*	-0.031**	-0.009
	(0.023)	(0.067)	(0.012)	(0.017)
CD4(301-350)	-0.037	-0.152*	-0.025*	-0.002
	(0.022)	(0.067)	(0.012)	(0.018)
CD4(351-400)	-0.027	-0.092	-0.013	0.003
	(0.023)	(0.067)	(0.012)	(0.018)
CD4(401-450)	-0.053*	-0.036	-0.032**	-0.019
	(0.023)	(0.068)	(0.012)	(0.018)
CD4(451-500)	-0.088***	-0.013	-0.005	0.031
	(0.025)	(0.075)	(0.013)	(0.020)
Treat $\times CD4(0-200)$	0.058**	0.102*	-0.033	-0.046*
	(0.022)	(0.044)	(0.017)	(0.020)
Treat $\times CD4(201-250)$	0.033	0.096	-0.073**	-0.103***
	(0.033)	(0.063)	(0.024)	(0.027)
Treat $\times CD4(251-300)$	-0.053	0.055	-0.083***	-0.105***
	(0.030)	(0.070)	(0.022)	(0.025)
Treat $\times CD4(301-350)$	-0.048	0.067	-0.059**	-0.082**
	(0.029)	(0.070)	(0.021)	(0.025)
Treat $\times CD4(351-400)$	-0.068*	-0.002	-0.084***	-0.100***
	(0.029)	(0.069)	(0.021)	(0.025)
Treat $\times CD4(401-450)$	0.055	0.038	0.030	0.018
	(0.029)	(0.071)	(0.021)	(0.025)
Treat $\times CD4(451-500)$	0.062*	-0.014	-0.022	-0.058*
	(0.030)	(0.077)	(0.022)	(0.026)
Constant	0.296***	0.280***	0.299***	0.327***
	(0.010)	(0.034)	(0.006)	(0.009)
Year dummies	Yes	Yes	Yes	Yes
R^2	0.09	0.09	0.07	0.07
Number of obs.	1,1829,338	1,1825,398	25,231,886	25,222,795

Notes: This table presents estimation results for the sample used for the back-of-the-envelope calculation. The estimation is performed as an OLS regression where the dependent variable is a dummy for employment. The sample used for column (1) Table C.1. Columns (1) and (2) correspond to the estimation where actual CD4 counts are used. Columns (3) and (4) correspond to the estimation where imputed CD4 counts are used. Columns (2) and (4) are estimated on a restricted sample where observations for the control cohorts are removed after 1995, when HAART treatment becomes available.